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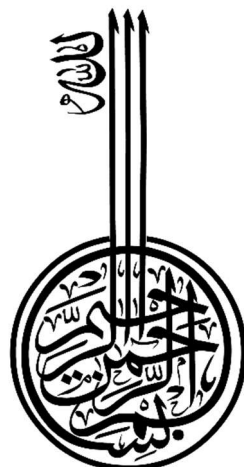
**Option: Sciences and Technologies of Information and
Communication (STIC)**

A Multimodal Deep Learning Platform for Predicting Non-Communicable Diseases Using ECG, EEG, MRI and Echocardiography Data

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Dedication



I dedicate this work to my mother, whose unwavering love and sacrifices made this journey possible; to my brothers and sisters, for their constant encouragement; and to all the teachers who, from primary school to the Master's level, transmitted to me a passion for learning. This thesis is the fruit of your trust.

Abstract

This thesis presents the design and implementation of a multimodal clinical decision-support platform that integrates four medical modalities – 12-lead Electrocardiogram (ECG), Electroencephalogram (EEG), Magnetic Resonance Imaging (MRI) and Echocardiography (ECHO) – under a single modular architecture for the prediction and assistance in the diagnosis of Non-Communicable Diseases. The platform integrates four pretrained deep-learning components (DenseNet-1D ensemble, U-Net, Vision Transformer ViT-B/16 and EchoNet-Dynamic) and a project-trained EEG IIC head built on the BIOT biosignal transformer. Two methodologically instructive engineering interventions are reported: per-pathology ECG threshold calibration that raised macro F1 from 0.514 to 0.711 on PTB-XL fold 10, and a double-sigmoid bug-fix that restored the U-Net Dice coefficient from 0.02 to 0.85 on the LGG dataset. The architecture follows a layered design with a Django REST Framework backend, a React 18 single-page frontend and a MongoDB document store. Doctor isolation and a uniform result-envelope contract guarantee partial-failure tolerance. The brain-heart synergy hypothesis that motivates multimodal integration is articulated as the principal future-work direction; the present multimodal claim is realised at the level of a rule-based PDF report rather than a learned cross-modal fusion.

Keywords: multimodal deep learning, ECG, EEG, MRI, echocardiography, brain-heart axis, non-communicable diseases.

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General Introduction



Project Background

Non-communicable diseases — cardiovascular pathologies, cancers, chronic respiratory disorders, and diabetes — account for approximately seventy-four per cent of global deaths and constitute the dominant burden on contemporary healthcare systems. Cardiovascular disease alone is responsible for nearly seventeen million deaths annually, and cancers for approximately ten million. Brain cancer, although representing only one to two per cent of all malignancies by incidence, carries a disproportionately high mortality and morbidity due to anatomical constraints on surgical resection and to the cognitive, motor, and autonomic impairments produced by tumor mass effect.

The artificial separation of cardiology and oncology is increasingly being challenged by an expanding body of physiological and clinical evidence demonstrating that the heart and the brain are coupled by anatomical, neural, humoral, and pharmacological pathways. Brain pathologies disrupt autonomic regulation and produce measurable electrocardiographic abnormalities; cancer treatments routinely affect cardiac function as well as their primary tumor target; and chronic cardiovascular dysfunction degrades cerebral perfusion. The emerging fields of cardio-oncology and neuro-cardiology formalize the integrative perspective that this thesis adopts as its scientific anchor.

Parallel to these clinical developments, artificial intelligence and particularly deep learning has reached clinically competitive performance across most medical signal and imaging modalities. Convolutional architectures match cardiologist-level accuracy on electrocardiogram classification, segment brain tumors on magnetic resonance imaging with expert agreement, classify electroencephalogram rhythms across pathological

categories, and quantify cardiac function from echocardiographic video. The natural research question is therefore whether the integration of these modalities under a unified deep learning framework can capture the brain-heart coupling and convert it into actionable clinical prediction.

Problem Statement

Despite the per-modality progress, three challenges remain unresolved.

First, no operational platform integrates electrocardiography, electroencephalography magnetic resonance imaging, and echocardiography for non-communicable disease prediction under a single architecture with a single clinician-facing interface. Each modality has its own pre-trained model ecosystem, its own preprocessing conventions, and its own evaluation benchmarks. The clinician's workflow consequently remains fragmented even when the underlying artificial-intelligence components are state-of-the-art.

Second, no public dataset provides synchronized, time-aligned recordings across all four target modalities for the same patient with documented non-communicable-disease status. This dataset alignment problem prevents the direct training of a multimodal classifier on paired multimodal data. It is the principal scientific obstacle to the empirical validation of the brain-heart synergy hypothesis.

Third, the explainability and clinical-trust requirements imposed by regulators and clinicians are not yet satisfied by the deep-learning systems most frequently reported in the literature. A platform intended for clinical decision support must integrate explainability throughout, not as an afterthought, and must report its limitations openly rather than rhetorically.

Proposed Solutions

This thesis proposes and implements a multimodal clinical decision support platform organized around five principles.

First, the platform integrates pre-trained, peer-reviewed deep learning models — a U-Net for brain tumor segmentation, a Vision Transformer for tumor classification, an ensemble of seven DenseNet-1D classifiers for arrhythmia detection, and NeuroKit2 for heart-rate-variability analysis — rather than networks trained from scratch. The

engineering of the integration is the research output for a Master's project of finite duration.

Second, the architecture is explicitly modality-agnostic. Each medical modality is encapsulated within an independent pipeline module that exposes a uniform result-envelope contract. The addition of a new modality requires only the implementation of a new module without modification of the authentication, persistence, or reporting layers.

Third, the platform addresses the dataset alignment problem head-on. Each modality is validated on its native public benchmark (PTB-XL for the electrocardiogram, LGG and the Brain Tumor MRI Dataset for the magnetic resonance imaging, and the analogous benchmarks for the future electroencephalography and echocardiography integrations). The brain-heart synergy hypothesis test, which requires paired multimodal data, is positioned as the principal long-term contribution that the present platform makes possible: a virtual multimodal cohort, constructed by transparent cross-cohort demographic and disease-stage matching, will host a learned attention-based fusion module trained to detect cross-modal correlations consistent with the brain-heart axis literature.

Fourth, the platform delivers a complete, deployable software system: a Django REST Framework backend, a React-based clinician-facing frontend, a MongoDB document store, secure JSON-Web-Token-based authentication, and a ReportLab-based combined PDF report. The combined report aggregates per-modality findings into a clinician-readable document and includes a rule-based combined interpretation that contextualizes the multimodal findings against the brain-heart axis literature. The interpretation is template-based, not learned; the scope of this multimodal claim is reported openly throughout.

Fifth, the platform reports two methodologically instructive implementation discoveries. The calibration of per-pathology decision thresholds for the electrocardiogram ensemble raised the macro F1 score from 0.513 to 0.711 on PTB-XL fold ten with no retraining. The diagnosis and removal of a redundant sigmoid in the segmentation pipeline raised the Dice coefficient from 0.02 to 0.85 on the LGG MRI Segmentation dataset.

Document Plan

The thesis is organized into three chapters, framed by this general introduction and a general conclusion.

Chapter 1, State of the Art, surveys the scientific and methodological context. It reviews artificial intelligence in healthcare, the epidemiology and clinical reality of non-communicable diseases, the pathophysiology of the brain-heart axis, the deep learning methods applied to each of the four target modalities, the multimodal-learning paradigms, the explainable-artificial-intelligence techniques, and the related work.

Chapter 2, Methods and Tools, specifies the platform itself. It documents the system architecture, the public datasets that feed it, the modality-agnostic preprocessing principles, the four modality pipelines (electrocardiography and magnetic resonance imaging implemented, electroencephalography and echocardiography designed-but-deferred), the feature extraction conventions, the multimodal fusion strategy, the deep learning models employed, the explainability module, and the backend, frontend, and database stacks.

Chapter 3, Results and Discussion, reports the experimental evaluation of the three implemented components on their public benchmarks. It documents the two methodologically instructive implementation discoveries (threshold calibration, double-sigmoid bug), compares the platform's numbers with published baselines, discusses the implications for the brain-heart synergy hypothesis, and lists the limitations and future improvements that follow directly from them.

The general conclusion synthesizes the contribution of the thesis as a whole and articulates the research agenda for the next phase, in which the brain-heart synergy hypothesis is the central object of investigation and the deferred modalities (electroencephalography and echocardiography) are the immediate engineering deliverables.

State of the Art

Introduction

Non-communicable diseases (NCDs) impose the dominant burden on global mortality and healthcare systems, with cardiovascular disease and cancer occupying the first two ranks worldwide. Although clinical practice has historically partitioned the management of these conditions by organ system, an expanding body of physiological and clinical evidence demonstrates that the heart and the brain are coupled by anatomical, neural, humoral, and pharmacological pathways.

In parallel, artificial intelligence (AI), and particularly deep learning, has reached clinically competitive performance across most medical signal and imaging modalities. Convolutional architectures match cardiologist-level accuracy on electrocardiogram (ECG) classification, segment brain tumors on magnetic resonance imaging (MRI) with expert agreement, classify electroencephalogram (EEG) rhythms across pathological categories, and quantify cardiac function from echocardiographic video.

1.1 Artificial Intelligence in Healthcare

The convergence of large-scale biomedical data, graphics processing units, and deep neural network architectures has transformed medical informatics over the past decade [1]. Convolutional neural networks (CNNs) have matched dermatologist performance in skin cancer classification [2], radiologist performance in pneumonia detection from chest radiographs, and ophthalmologist performance in diabetic retinopathy grading [3].

Three challenges remain central to the field. First, data heterogeneity across institutions, devices, and acquisition protocols undermines model generalization.

From an architectural perspective, contemporary medical AI systems share a common organization. A data ingestion layer handles raw inputs from clinical devices and electronic health records, applying format conversion and quality control.

1.2 Non-Communicable Diseases

Non-communicable diseases comprise cardiovascular disease, cancer, chronic respiratory disease, diabetes, and certain neurological conditions. According to the World Health Organization, NCDs account for approximately seventy-four per cent of global deaths and eighty per cent of premature mortality below the age of seventy [5].

Cardiovascular disease encompasses ischemic heart disease, cerebrovascular disease, heart failure, cardiomyopathies, valvular disorders, and rhythm abnormalities. The principal modifiable risk factors are hypertension, dyslipidemia, smoking, obesity, physical inactivity, and diabetes.

Cancer is biologically heterogeneous but unified by uncontrolled cellular proliferation, evasion of apoptosis, and capacity for invasion. Brain cancer, although representing only one to two per cent of all malignancies by incidence, carries disproportionate morbidity and mortality.

The clinical and epidemiological literature increasingly recognizes that NCDs do not act in isolation. Patients with brain tumors experience markedly elevated rates of cardiovascular events, partly through shared risk factors such as age, partly through treatment-induced cardiotoxicity, and partly through direct neuroanatomical disruption of cardiac autonomic regulation [6].

1.3 The Brain-Heart Axis

The brain-heart axis denotes the bidirectional anatomical and physiological coupling between the central nervous system and the cardiovascular system, mediated primarily by the autonomic nervous system. This subsection establishes the pathophysiological basis on which the present work builds its multimodal predictive hypothesis.

1.3.1 The central autonomic network

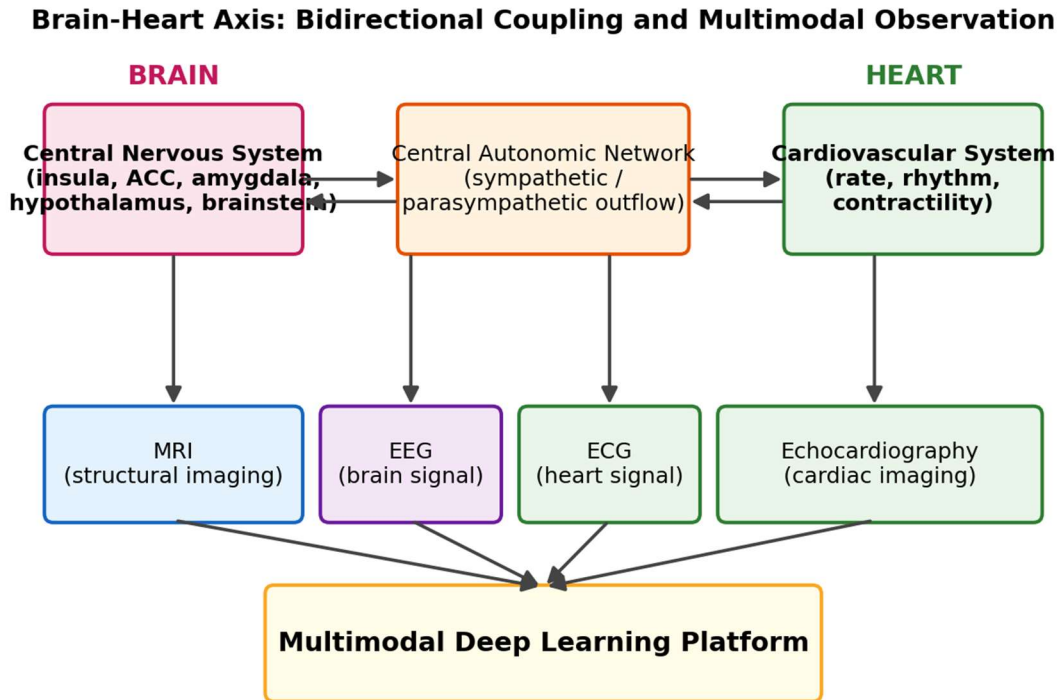


Figure 1: Schematic of the brain-heart axis. Bidirectional autonomic coupling between the central nervous system and the cardiovascular system is observed through four medical modalities (ECG, EEG, MRI, echocardiography) that the proposed platform integrates.

The central autonomic network (CAN) comprises a distributed set of cortical and subcortical structures that regulate sympathetic and parasympathetic outflow to the heart, vasculature, and other visceral organs [8]. Key nodes include the insular cortex, the anterior cingulate cortex, the amygdala, the hypothalamus, the periaqueductal gray, the nucleus of the solitary tract, the rostral ventrolateral medulla, and the nucleus ambiguus.

1.3.2 Neurogenic cardiac signatures

Disruption of the central autonomic network – through stroke, traumatic brain injury, subarachnoid hemorrhage, or tumor – produces a recognized spectrum of electrocardiographic abnormalities. These include corrected QT interval prolongation, T-wave inversion, ST-segment elevation or depression mimicking myocardial ischemia, U-wave appearance, and arrhythmias ranging from sinus bradycardia to polymorphic ventricular tachycardia [9].

1.3.3 Brain tumors and autonomic dysfunction

Brain tumors affect autonomic function through three mechanisms. The first is direct invasion or compression of central autonomic network structures, observed most consistently with tumors of the insular cortex, brainstem, hypothalamus, or third ventricle.

1.3.4 Cardio-oncology

Beyond direct neurogenic effects, brain cancer treatment imposes a substantial cardiovascular burden. Anthracycline chemotherapy produces dose-dependent cardiomyopathy through mitochondrial iron accumulation and oxidative stress.

1.3.5 Implications for the present work

Three implications structure the methodology developed in subsequent chapters. First, the heart and brain are not independent organs, and treating them as separate prediction targets is biologically incorrect.

1.4 Modality-Specific Deep Learning

This section reviews the deep learning methods developed for each of the four modalities employed in the present work: electrocardiography, electroencephalography, magnetic resonance imaging, and echocardiography.

1.4.1 Electrocardiography

The electrocardiogram records the electrical depolarization and repolarization cycle of the myocardium through electrodes placed on the body surface. The standard twelve-lead clinical ECG samples at two hundred fifty to five hundred hertz over ten seconds, while ambulatory and wearable ECG may sample continuously over days.

Deep learning approaches treat the ECG as a one-dimensional time series. The principal architectures include:

- ▶ **One-dimensional convolutional neural networks (1D-CNN):** stacks of one-dimensional convolution and pooling layers that learn local waveform features at multiple temporal scales [10].
- ▶ **Residual networks adapted to one-dimensional input (ResNet1D):** deep residual networks enabling training of architectures with dozens of layers while

preserving gradient flow through skip connections [11]. Hannun et al. trained a thirty-four layer ResNet1D on more than ninety thousand single-lead ambulatory ECG recordings and reported cardiologist-level rhythm classification across twelve diagnostic classes [12].

- **Recurrent and attention-based models: long short-term memory networks and Transformer architectures applied to ECG capture longer-range temporal dependencies relevant to arrhythmia detection.**

Formally, a one-dimensional convolution layer applied to an ECG signal of length T produces an output feature map according to:

$$y[t] = \sigma \left(\sum_{k=0}^{K-1} w[k] \cdot x[t+k] + b \right)$$

where w is the learned kernel of length K , b is a bias term, and σ is a non-linear activation function such as the rectified linear unit. Stacks of such layers, interleaved with pooling operations, progressively widen the receptive field over the temporal axis until the entire heartbeat is encoded.

Public datasets relevant to the present work include the PhysioNet MIT-BIH Arrhythmia Database [13], comprising forty-eight half-hour two-lead recordings with beat-level annotations; the PTB-XL database [14], with twenty-one thousand eight hundred thirty-seven clinical twelve-lead ECGs annotated according to seventy-one diagnostic statements; and the Icentia11k corpus for ambulatory analysis. Limitations of ECG deep learning include sensitivity to lead placement, susceptibility to motion artefacts, and pronounced domain shift between clinical and ambulatory recordings.

1.4.2 Electroencephalography

Electroencephalography measures the summed postsynaptic potentials of cortical pyramidal neurons through scalp electrodes, typically arranged according to the ten-twenty international system [15]. Clinical EEG samples at two hundred fifty to one thousand hertz across nineteen to two hundred fifty-six channels.

Deep learning for EEG must contend with low signal-to-noise ratio, pronounced inter-subject variability, and a comparatively low spatial resolution. The principal architectures include:

- **EEGNet [16]: a compact convolutional neural network designed specifically for EEG, employing depthwise and separable convolutions to learn spatial filters per frequency band. EEGNet has become a de facto baseline across motor imagery, P300, error-related negativity, and seizure detection tasks.**

- ▶ **ChronoNet and recurrent EEG models:** architectures that combine multiscale convolutional feature extraction with bidirectional gated recurrent units for long-recording analysis [17].
- ▶ **Transformer-based EEG models:** recent work applies self-attention across electrode-time tokens to capture global spatiotemporal dependencies.

Public benchmarks include the Temple University Hospital EEG Corpus [18], the largest clinical EEG dataset with more than thirty thousand recordings; the CHB-MIT Scalp EEG Database [19], a paediatric seizure benchmark; and the PhysioNet EEG Motor Movement and Imagery Dataset [20] for brain-computer interface tasks.

For the present work, EEG provides direct access to cortical activity in regions implicated in autonomic regulation, complementing both ECG-derived autonomic markers and MRI-derived structural information. EEG abnormalities are also a recognized feature of brain tumors, with focal slowing in the delta and theta ranges over the tumor region often preceding the appearance of clinical seizures.

1.4.3 Magnetic Resonance Imaging

Magnetic resonance imaging generates high-resolution volumetric images of soft tissue by exploiting the relaxation behaviour of hydrogen nuclei in a strong magnetic field [21]. Brain tumor imaging typically combines four sequences: native T1-weighted, post-gadolinium T1-weighted, T2-weighted, and Fluid-Attenuated Inversion Recovery.

Deep learning for brain tumor MRI is dominated by encoder-decoder segmentation architectures:

- ▶ **U-Net [22]:** the canonical biomedical segmentation network, with a contracting encoder and a symmetric expanding decoder linked by skip connections. The skip connections preserve spatial information lost in the bottleneck and have become a defining structural motif of the field.
- ▶ **Three-dimensional U-Net and V-Net [23]:** volumetric extensions that operate directly on three-dimensional MRI volumes rather than two-dimensional slices, exploiting inter-slice context.
- ▶ **nnU-Net [24]:** a self-configuring U-Net pipeline that automatically adapts preprocessing, network topology, and training schedule to each dataset. This pipeline has dominated the Brain Tumor Segmentation Challenge leaderboard for several consecutive editions.

- ▶ **Transformer-based segmentation models:** architectures such as Swin UNETR [25] apply hierarchical self-attention to three-dimensional volumes, achieving competitive performance on multimodal brain tumor segmentation.

The principal benchmark dataset is the Brain Tumor Segmentation Challenge (BraTS) [26], which provides multimodal MRI volumes with expert-annotated tumor sub-regions – necrotic core, enhancing tumor, and peritumoral edema – for several thousand glioma patients across successive editions. Additional resources include the Cancer Genome Atlas (TCGA) and the Cancer Imaging Archive (TCIA) for clinical and genomic correlates.

For the brain-heart synergy hypothesis investigated in this thesis, MRI provides the unique ability to localize tumors with respect to autonomic network nodes. Probabilistic atlases of the insular cortex, brainstem nuclei, and hypothalamus can be registered to patient volumes to compute the proportion of each tumor that overlaps with central autonomic regions, generating an anatomically meaningful covariate for correlation with ECG-derived autonomic markers.

1.4.4 Echocardiography

Echocardiography uses high-frequency ultrasound to generate real-time images of cardiac structure and function. Standard transthoracic echocardiography acquires multiple views – parasternal long-axis, parasternal short-axis, apical four-chamber, apical two-chamber, and subcostal – as cine loops spanning several cardiac cycles.

Deep learning for echocardiography presents two distinctive challenges relative to other imaging modalities: the data are spatiotemporal video rather than static images, and the acquisition is highly operator-dependent, with substantial variability in image quality and view angle. Representative architectures include:

- ▶ **EchoNet-Dynamic [27]:** a spatiotemporal convolutional network trained to estimate left ventricular ejection fraction directly from apical four-chamber cine loops, achieving expert-level agreement on more than ten thousand echocardiograms.
- ▶ **R(2+1)D and three-dimensional convolutional video models:** architectures that factorize spatial and temporal convolutions for efficient processing of echocardiographic video.

- ▶ **View classification and segmentation pipelines:** upstream models that automatically identify the view being acquired and segment the cardiac chambers, providing a structured input for downstream functional analysis.

The principal public dataset is EchoNet-Dynamic [27], comprising ten thousand thirty labelled apical four-chamber echocardiographic videos with frame-level segmentation of the left ventricle and clip-level ejection fraction labels. The CAMUS dataset [28] provides annotated end-diastolic and end-systolic frames for segmentation.

1.5 Multimodal Deep Learning

Multimodal learning combines information from heterogeneous sources into a unified representation suitable for prediction, with the assumption that complementary signals yield strictly more information than the sum of their parts. Three principal fusion paradigms are recognized:

- ▶ **Early fusion:** modality-specific raw or low-level features are concatenated at the input of a single network. This approach is effective only when modalities share a common representation space and a common sampling rate.
- ▶ **Late fusion:** modality-specific networks are trained independently and their outputs — logits or class probabilities — are combined by averaging, weighted voting, or a meta-learner. Late fusion is robust to missing modalities but loses inter-modal interactions because each network never sees the others.
- ▶ **Hybrid or intermediate fusion:** modality-specific encoders produce mid-level embeddings which are then combined through concatenation, attention, or a Transformer block before a shared prediction head. Most modern multimodal medical models adopt this scheme.

Attention-based fusion allows the model to weight modalities differentially per patient, increasing robustness to noisy or missing channels [30]. Cross-modal Transformers extend this principle by computing self-attention across tokens drawn from multiple modalities within a single sequence.

A separate strand of work addresses learning multimodal representations without paired training data. Contrastive learning — exemplified by CLIP in vision-language [31] and extended to medical signal pairs — learns a shared latent space in which paired samples are pulled together and unpaired samples are pushed apart.

The principal open problem in multimodal medical learning is the dataset alignment problem. With rare exceptions, large-scale public datasets contain a single modality, and

few cohorts provide simultaneous, time-synchronized recordings across all four modalities of interest in the present work.

1.6 Explainable Artificial Intelligence

Clinical adoption of deep learning models depends critically on explainability. A clinician asked to act on an algorithmic prediction requires a justification that can be audited, communicated to the patient, and integrated into the documented chain of clinical reasoning.

Post-hoc local explainers. SHAP, the Shapley Additive Explanations framework, assigns each input feature an additive contribution to the model output, derived from cooperative game theory.

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! \cdot (|N| - |S| - 1)!}{|N|!} \cdot [f(S \cup \{i\}) - f(S)]$$

where N is the full set of features and S ranges over coalitions of features not containing i . SHAP values satisfy theoretical properties of efficiency, symmetry, and additivity that ground their interpretation.

Gradient-based saliency. Gradient-weighted Class Activation Mapping (Grad-CAM) [33] visualizes class-discriminative regions in convolutional inputs by weighting feature maps with gradients of the target class score.

Intrinsically interpretable architectures. Attention weights [34], prototype networks, and concept-bottleneck models offer interpretability built into the architecture itself rather than imposed afterwards.

For ECG, SHAP has been applied at the beat and segment level to identify QRS or T-wave regions driving arrhythmia predictions. For MRI, Grad-CAM and attention maps highlight tumor and peritumoral regions [35].

Regulatory bodies, including the European Medicines Agency and the United States Food and Drug Administration, increasingly require explainability evidence as part of clinical artificial intelligence submissions [36]. The General Data Protection Regulation of the European Union and equivalent national frameworks formalize a right to explanation for individuals subject to automated decision-making.

1.7 Related Work

This subsection reviews prior work along three axes relevant to the present contribution: computational studies of the brain-heart axis, multimodal medical artificial intelligence platforms, and cross-cohort learning strategies for the dataset alignment problem.

1.7.1 Computational studies of the brain-heart axis

Several research groups have demonstrated machine learning detection of brain pathology from cardiac signals. Heart rate variability features have been used to predict stroke outcome at thirty and ninety days, to detect cognitive impairment in Parkinson's disease, and to predict one-year mortality in patients with high-grade glioma.

1.7.2 Multimodal medical artificial intelligence platforms

Recent multimodal frameworks include MedCLIP [40] and BiomedCLIP in the image-text domain, and the MultiBench benchmark suite for multimodal learning more generally [41]. In the context of multimodal physiological signals, work has been reported on the integration of ECG and photoplethysmography for cardiovascular risk stratification [42] and of EEG and ECG for automated sleep staging [43].

1.7.3 Cross-cohort learning and the dataset alignment problem

Federated learning enables joint model training across institutions without sharing raw patient data, addressing both data scarcity and privacy concerns. Domain adaptation methods reduce performance degradation caused by distribution shift between cohorts.

1.8 Synthesis and Discussion

The reviewed literature establishes four anchor points for the present work. First, deep learning has reached clinically competitive performance on each of the four modalities considered in this thesis, evaluated on dedicated public benchmarks.

Two consequences follow for the design of this thesis. The first is that any contribution claiming to test the brain-heart synergy hypothesis must explicitly confront the dataset alignment problem, rather than circumvent it by silently combining unrelated cohorts.

Conclusion

This chapter has surveyed the state of the art across four medical modalities, the deep learning architectures applied to each of them, multimodal fusion strategies, explainability techniques, and the pathophysiology of the brain-heart axis.

Methods and Tools

Introduction

Chapter 1 established the scientific and methodological landscape within which this thesis operates. It surveyed the four medical modalities targeted by the platform, examined the deep learning architectures applied to each, reviewed the pathophysiology of the brain-heart axis, and identified the dataset alignment problem as the principal obstacle to direct empirical investigation of cross-modal interactions in non-communicable diseases.

The chapter is organized around an explicit principle of engineering honesty. Of the four target modalities, two are fully implemented in the present version of the platform – magnetic resonance imaging (MRI) and twelve-lead electrocardiography (ECG) – while two are designed for under the modular architecture but not yet integrated – electroencephalography (EEG) and echocardiography (ECHO).

The remainder of the chapter is organized as follows. Section 2.1 presents the system architecture and its governing design principles.

2.1 System Architecture

The platform follows a layered modular architecture inspired by clean architecture principles and the representational state transfer style for the application programming interface [32]. Four logical layers communicate through narrow contracts: a single-page web application presents data to the clinician; a Django REST Framework service exposes typed endpoints and orchestrates inference; a domain layer hosts the inference engine

and modality-specific pipelines; and a document-oriented MongoDB database persists patient records and analysis artefacts.

2.1.1 Five governing design principles

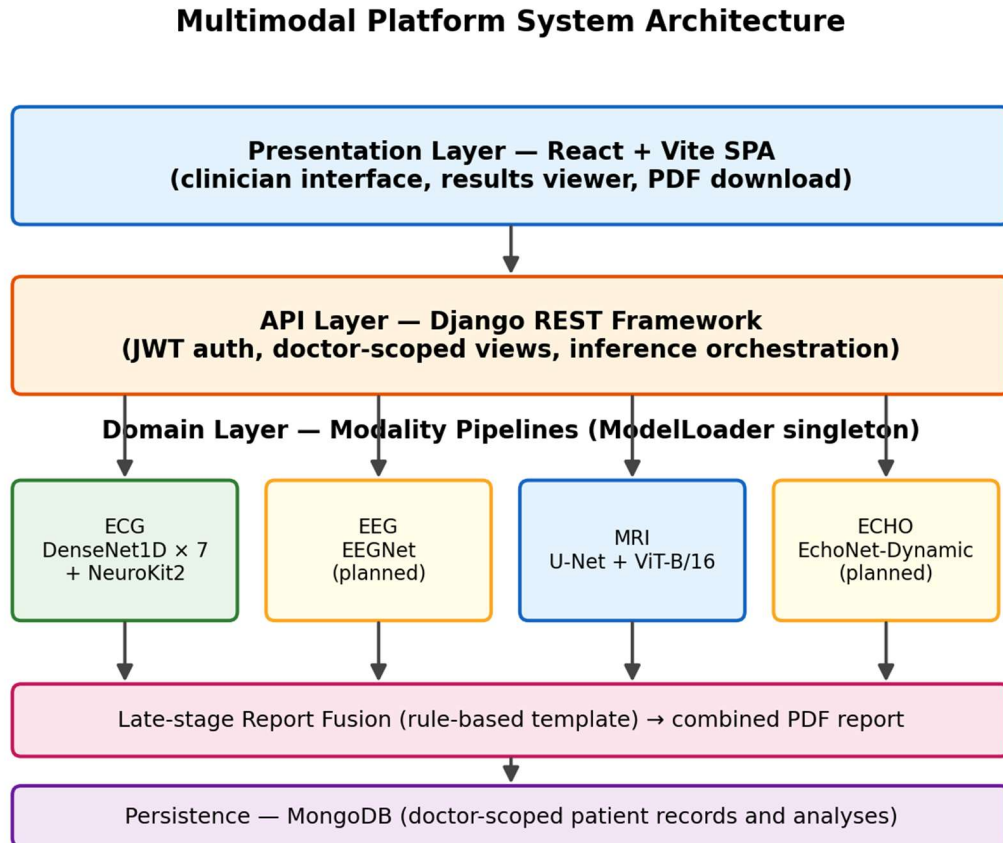


Figure 2: System architecture of the multimodal platform. A layered design separates the presentation, API, domain (per-modality pipelines) and persistence layers; the result-envelope contract ensures partial-failure tolerance and modality independence.

The architecture is governed by five principles, each with both a scientific and a software-engineering justification.

Modular pipeline isolation. Each medical modality is encapsulated in a self-contained pipeline module that exposes a uniform interface.

Synchronous inference with bounded latency. Inference is executed in the request thread under a five-minute timeout.

Lazy singleton model loading. Pretrained model weights are large — the Vision Transformer alone occupies approximately three hundred fifty megabytes.

Doctor-scoped data isolation. Every database query is parameterized by the identity of the requesting clinician, extracted from the JSON Web Token carried in the Authorization header.

Result envelope contract. Every pipeline returns a Python dictionary of the form {status: string,...fields, error?: string}.

2.2 Data Acquisition

The platform is developed and validated entirely on publicly available, peer-reviewed benchmark datasets.

2.2.1 ECG dataset

Table 1: Public datasets used or planned for each modality.

Dataset	Modality	Size	Use in this work	Reference
PTB-XL v1.0.3	ECG (12-lead)	21,837 records	ECG ensemble validation (fold 10)	PhysioNet
Brain Tumor MRI Dataset	MRI (T1-weighted)	7,022 images, 4 classes	ViT-B/16 classification testing	Kaggle (Nickparvar)
LGG MRI Segmentation (TCGA)	MRI (FLAIR)	3,929 slices, 110 patients	U-Net segmentation evaluation	Buda et al., 2019
TUH EEG Corpus (planned)	EEG (multi-channel)	~30,000 recordings	EEGNet evaluation (future work)	Obeid & Picone, 2016
EchoNet-Dynamic (planned)	Echocardiography (video)	10,030 studies	EchoNet-Dynamic evaluation (future work)	Ouyang et al., 2020

The ECG pipeline is validated on the PTB-XL database [14], version 1.0.3, distributed through the PhysioNet platform [37]. PTB-XL contains 21,837 twelve-lead ECG records of ten-second duration sampled at five hundred hertz, with seventy-one diagnostic statements annotated by cardiologists according to the SCP-ECG standard.

2.2.2 MRI datasets

The MRI pipeline employs two distinct public datasets reflecting two complementary tasks. For the segmentation task, the Lower Grade Glioma MRI Segmentation dataset of Buda et al. [39] is used: 3,929 two-dimensional FLAIR axial slices from 110 patients in The Cancer Genome Atlas, of which 1,373 contain tumor and 2,556 are tumor-free.

2.3 Data Preprocessing

Preprocessing transforms raw medical data into the canonical numerical representation expected by the deep learning models. The platform applies a common pattern across modalities — format normalization, resampling to a fixed shape, denoising, and standardization — with parameters specific to each modality.

Input-format permissiveness. Each pipeline accepts every realistic input format for its modality and converts internally to a canonical numerical array.

Modality-appropriate denoising. Each modality has a distinct noise spectrum and requires distinct treatment.

Standardization to the training distribution. Each pipeline applies the normalization scheme on which the corresponding model was trained, in order to avoid covariate shift between training and inference.

2.4 ECG Pipeline

The ECG pipeline screens for seven cardiac pathologies and computes time-domain heart rate variability metrics.

2.4.1 ECG preprocessing

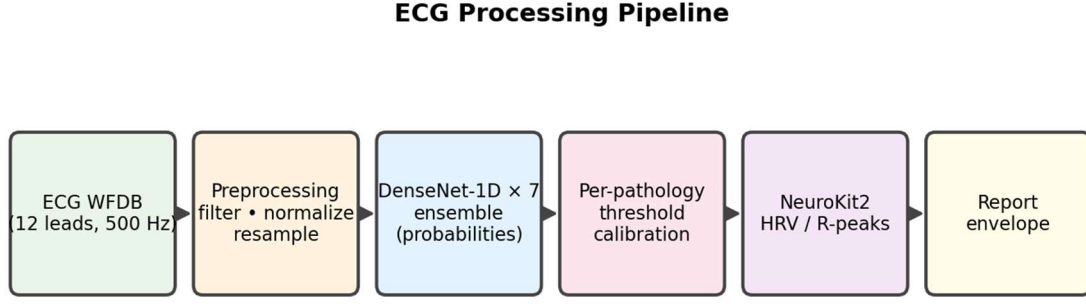


Figure 3: ECG processing pipeline: raw WFDB record → preprocessing → DenseNet-1D ensemble → per-pathology threshold calibration → NeuroKit2 HRV analysis → structured result envelope.

Three stages standardize arbitrary input recordings to the fixed numerical representation expected by the downstream models.

Stage one normalizes shape and sampling rate. Inputs are converted to a NumPy array of shape (twelve leads, five thousand samples) corresponding to a ten-second window at five hundred hertz.

Stage two suppresses non-cardiac contamination. A fourth-order Butterworth band-pass filter with cutoff frequencies of 0.5 Hz and 40 Hz is applied in the zero-phase forward-backward mode using SciPy's `filtfilt` function [43].

$$H(z) = \left(\sum_k b_k z^{-k} \right) / \left(1 + \sum_k a_k z^{-k} \right)$$

where the coefficients a_k and b_k are determined from the analog Butterworth prototype via the bilinear transformation. The zero-phase response is obtained by applying the filter forward, time-reversing the output, applying the filter again, and time-reversing once more — eliminating phase distortion at the cost of an effective filter order that is twice the nominal order.

Stage three normalizes amplitude per lead via z-score normalization:

$$\tilde{x}_l[t] = (x_l[t] - \mu_l) / (\sigma_l + \varepsilon)$$

where μ_l and σ_l are the mean and standard deviation of lead l over its five thousand samples, $\varepsilon = 10^{-8}$ is a small constant added to avoid division by zero in the case of a flat lead, and $\tilde{x}_l$ is the normalized lead.

2.4.2 ECG deep learning component

The preprocessed twelve-by-five-thousand tensor is forwarded to an ensemble of seven DenseNet-1D-121 binary classifiers [44], drawn from the ecglib library released by the Institute for System Programming of the Russian Academy of Sciences [41]. Each classifier targets one of the seven pathologies listed in Section 2.2.1.

2.4.3 ECG classical signal processing component

In parallel with the deep ensemble, NeuroKit2 [42] computes time-domain heart rate variability indices from lead II using a Pan-Tompkins-style R-peak detection algorithm. The indices computed are mean heart rate, the root mean square of successive RR-interval differences (RMSSD), the standard deviation of normal-to-normal intervals (SDNN), and the percentage of successive RR intervals differing by more than fifty milliseconds (pNN50).

2.5 EEG Pipeline

The EEG pipeline implements end-to-end Ictal-Interictal-Injury-Continuum (IIIC) classification using the BIOT biosignal transformer as a frozen encoder and a project-trained classification head. It is the only fully fine-tuned component of the platform.

2.5.1 EEG preprocessing

Raw EDF recordings are parsed and converted to a 16-channel longitudinal-bipolar montage, resampled to 200 Hz, and split into non-overlapping 10-second segments (2000 samples per segment). Each channel is normalised by its 95th-percentile amplitude, exactly reproducing the preprocessing rule used during BIOT pre-training and project fine-tuning to guarantee train/inference parity.

2.5.2 BIOT encoder and IIIC head

BIOT is a self-supervised transformer pretrained on five million MGH resting-EEG samples. The project vendors the BIOT source code, loads its public weights with a byte-exact key match (0 missing / 0 unexpected parameters), and freezes the encoder. A new ClassificationHead is appended and fine-tuned on the public Kaggle Harmful Brain Activity (HMS) dataset via `tools/train_eeg_head.py`.

2.5.3 Inference output

For each 10-second segment the head outputs probabilities over six IIC classes: SZ (Seizure), LPD (Lateralized Periodic Discharges), GPD (Generalized Periodic Discharges), LRDA (Lateralized Rhythmic Delta Activity), GRDA (Generalized Rhythmic Delta Activity), Other. Per-recording outputs aggregate to a class distribution, a dominant pattern and a harmful-activity flag (any segment classified SZ/LPD/GPD).

The EEG pipeline is designed under the same modular contract as the ECG and MRI pipelines but is not yet integrated in the present version of the platform.

2.6 MRI Pipeline

The MRI pipeline performs two complementary tasks: pixel-level segmentation of the tumor region and four-class classification of the tumor type.

2.6.1 MRI preprocessing

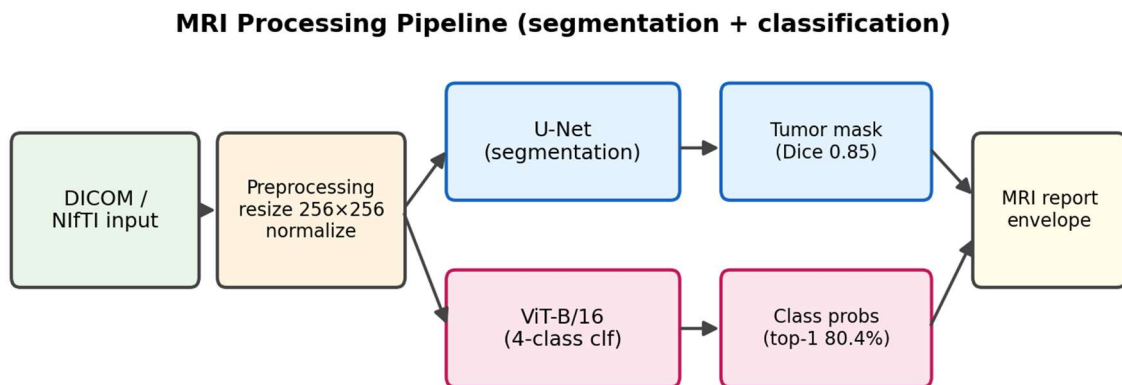


Figure 4: MRI processing pipeline: DICOM/NIfTI input is preprocessed once and routed in parallel to a U-Net segmentation branch (Dice 0.85 on LGG) and a Vision-Transformer classification branch (top-1 80.4% on Brain Tumor MRI Dataset).

Inputs are accepted in six formats (PNG, JPEG, TIFF, BMP, DICOM, NIfTI) and converted internally to RGB arrays.

The U-Net segmentation branch resizes the image to 256 by 256 pixels using bilinear interpolation and applies per-channel z-score normalization. The resulting tensor has shape (1, 3, 256, 256) and is consumed directly by the U-Net.

The Vision Transformer classification branch uses the HuggingFace AutoImageProcessor associated with the Devarshi model card, which performs the

appropriate resize and ImageNet-style normalization [45]. When a tumor bounding box is available from the segmentation branch, the image is cropped to that bounding box before classification; otherwise the full image is used.

2.6.2 MRI segmentation: U-Net

MRI Two-Stage Pipeline: Segmentation → Saturation Guard → Crop-and-Classify

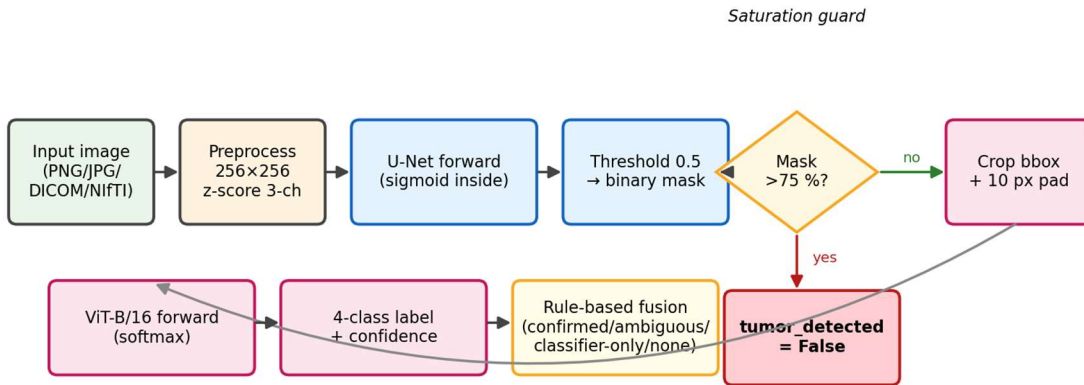


Figure 5: MRI two-stage pipeline. Segmentation runs first; a saturation guard rejects masks covering more than 75 percent of the image (degenerate output). When a tumour is detected, the image is cropped to the mask bounding box with 10-pixel padding and passed to the Vision Transformer; a rule-based fusion step produces one of four recommendations.

The segmentation network is the canonical U-Net of Ronneberger et al. [22], specifically the variant trained by Buda et al. [39] on the TCGA-LGG dataset and distributed via torch.hub under mateuszbuda/brain-segmentation-pytorch. The architecture comprises a contracting encoder path of four convolutional blocks with two-by-two max-pooling, a bottleneck, and a symmetric expanding decoder path with four transposed convolution blocks.

2.6.3 MRI classification: Vision Transformer

Tumor type classification is performed by a Vision Transformer (ViT-B/16) [46], an architecture that adapts the Transformer self-attention mechanism [47] to image recognition by treating sixteen-by-sixteen pixel patches as input tokens. The model used in the platform is a fine-tuned checkpoint distributed on the HuggingFace Hub under Devarshi/Brain_Tumor_Classification, with approximately 86 million parameters.

2.7 Echocardiography Pipeline

The echocardiography pipeline implements left-ventricular ejection-fraction (EF) regression and left-ventricle (LV) segmentation using the EchoNet-Dynamic models published by Ouyang et al. Both components are integrated as-is; weights are downloaded separately and the pipeline fails with a clear `FileNotFoundError` until they are provided.

2.7.1 Echo preprocessing

Echocardiographic video (.avi/.mp4) is decoded to 112x112 RGB frames and normalised using the EchoNet training statistics. Up to four sliding clips of 32 frames each are sampled per study; ejection fraction is averaged over the clip predictions and clamped to the physiological range [0, 100] percent.

2.7.2 EF regression and LV segmentation

Ejection fraction is regressed by an R(2+1)D-18 spatiotemporal convolutional network. Per-frame LV segmentation is produced by DeepLabV3 with a ResNet-50 backbone. End-diastole is defined as the frame of largest LV area and end-systole as the frame of smallest non-zero LV area. The continuous EF value is finally mapped to a clinical category (Reduced/HFrEF below 40 percent, Mildly reduced below 50 percent, otherwise Normal).

The echocardiography pipeline is, like the EEG pipeline, designed under the modular contract but not yet integrated.

2.8 Feature Extraction

Each modality contributes a structured feature vector to the downstream multimodal report.

2.8.1 ECG features

The ECG pipeline produces a seven-dimensional pathology probability vector from the DenseNet-1D ensemble, a four-dimensional heart-rate-variability vector from NeuroKit2 (mean HR, RMSSD, SDNN, pNN50), and two binary clinical flags (bradycardia, tachycardia) from the rule-based cross-check. The pathology probabilities are continuous

in the unit interval; the heart-rate-variability vector takes values in beats-per-minute or milliseconds depending on the index; and the clinical flags are binary.

2.8.2 MRI features

The MRI pipeline produces a four-dimensional tumor-class probability vector from the Vision Transformer, a binary segmentation mask from the U-Net, and a small set of mask-derived structural features: total tumor area in pixels at native resolution, mask centroid coordinates, and a boolean indicator of whether the tumor occupies the central forty per cent of the image (used as a rough surrogate for deep-brain involvement). These structural features are computed by classical image-processing operations on the binary mask.

2.9 Multimodal Fusion

Multimodal fusion combines the per-modality feature vectors produced by Section 2.8 into a unified clinical output.

2.9.1 Fusion paradigms reviewed

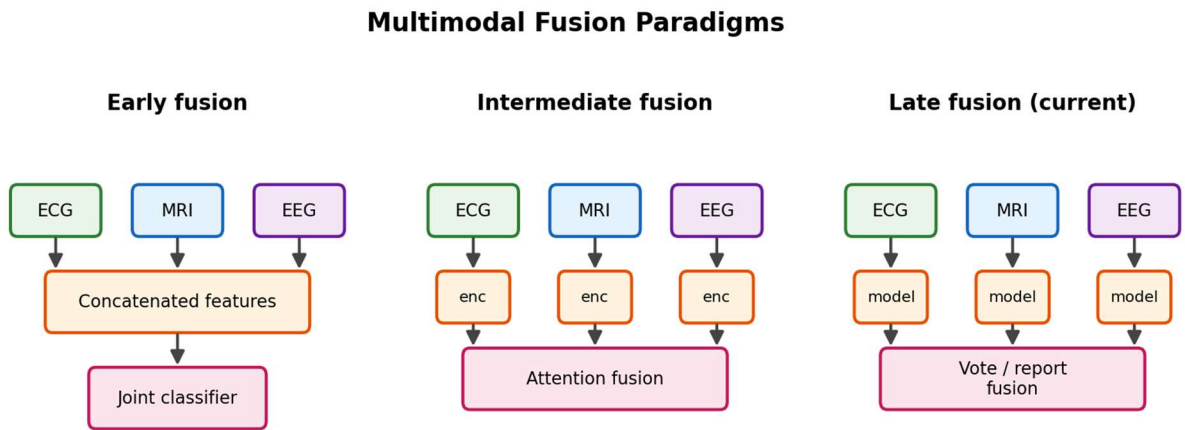


Figure 6: Multimodal fusion paradigms reviewed in this work. The current implementation realises late-stage report fusion; intermediate attention-based fusion is the principal future direction.

Three principal fusion paradigms are recognized in the multimodal medical artificial intelligence literature [29], [48]: early fusion, in which modality-specific raw or low-level features are concatenated at the input of a single network; late fusion, in which modality-specific networks are trained independently and their outputs are combined post-hoc; and hybrid or intermediate fusion, in which modality-specific encoders produce mid-

level embeddings which are then combined through concatenation, attention, or a Transformer block before a shared prediction head.

2.9.2 Currently implemented: late-stage report fusion

The current platform implements late-stage report fusion. The MRI and ECG pipelines run independently on their respective inputs and produce independent structured results.

2.9.3 Honest scope of the multimodal claim

The choice of late-stage report fusion is deliberate and follows from the dataset alignment problem articulated in Chapter 1. A learned intermediate fusion would require paired training data — synchronized MRI and ECG (and, in the full vision, EEG and ECHO) for the same patient with documented brain-cancer status — that does not exist publicly.

2.9.4 Future: learned intermediate fusion

The architecture explicitly admits a learned fusion module that consumes the four-modality embeddings produced by the individual pipelines and outputs a unified patient-level risk score. The candidate design is an attention-bottleneck Transformer that allows the model to weight modalities differentially per patient.

2.10 Deep Learning Models

Table 2: Deep learning components integrated by the platform.

Component	Architecture	Task	Status	Parameters
ECG ensemble	DenseNet-1D- 121 × 7	7-pathology classification	Implemented	~7 × 8 M
ECG signal proc.	NeuroKit2	R-peak / HRV	Implemented	—
MRI segmentation	U-Net (Buda et al.)	Tumor mask	Implemented	~7.8 M
MRI classification	ViT-B/16 (HuggingFace)	4-class tumor type	Implemented	~86 M
EEG	EEGNet	Rhythm / event classification	Planned	~2 k
Echocardiography	EchoNet- Dynamic (R(2+1)D)	EF estimation	Planned	~31 M

The platform integrates pretrained, peer-reviewed deep learning models rather than networks trained from scratch.

2.10.1 U-Net (MRI segmentation)

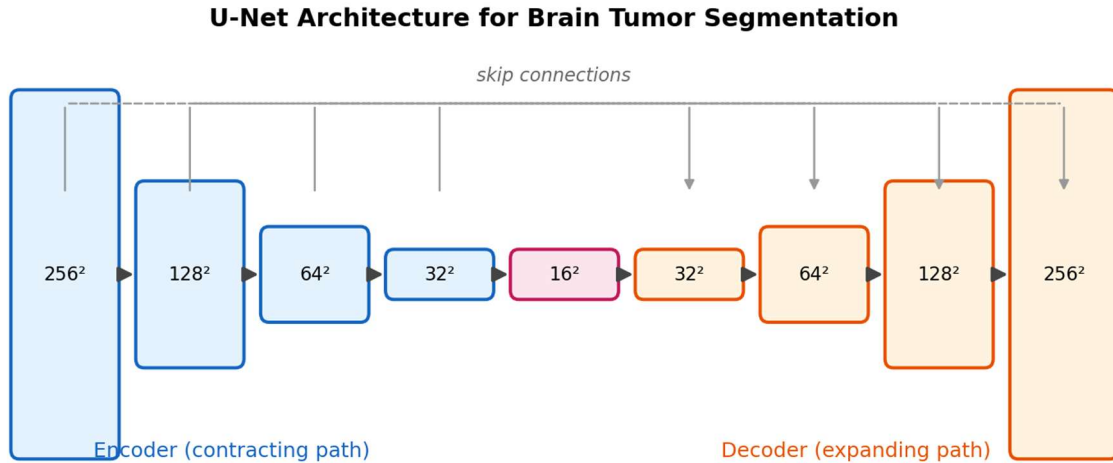


Figure 7: Simplified U-Net architecture: a contracting encoder of convolutional blocks captures context, an expanding decoder restores spatial resolution, and skip connections preserve fine-grained localisation information across resolutions.

The U-Net [22] follows an encoder-decoder topology with skip connections that preserve spatial information lost in the bottleneck. The variant used here is the open-source brain-segmentation-pytorch implementation of Buda et al. [39], pre-trained on the TCGA-LGG dataset and distributed via torch.hub.

2.10.2 Vision Transformer ViT-B/16 (MRI classification)

ViT-B/16 [46] adapts the Transformer self-attention mechanism [47] to image recognition by treating 16 by 16 pixel patches as input tokens. The platform uses a fine-tuned checkpoint distributed on the HuggingFace Hub under Devarshi/Brain_Tumor_Classification, with approximately 86 million parameters.

2.10.3 DenseNet-1D-121 ensemble (ECG)

The ECG ensemble comprises seven binary classifiers, each a DenseNet-121 [44] adapted to one-dimensional input as released by the ecglib toolkit [41]. Each classifier addresses one pathology drawn from the SCP-ECG vocabulary.

2.10.4 NeuroKit2 (ECG signal processing)

2.10.5 BIOT transformer (EEG)

The Biosignal Transformer (BIOT) is a Transformer encoder pretrained self-supervised on five million MGH resting-EEG samples. The platform uses BIOT as a frozen feature extractor; a ClassificationHead is added on top and trained in-repo on the Kaggle HMS dataset to produce the IIC six-class distribution. This is the only deep-learning model whose weights were modified within the present work.

2.10.6 EchoNet-Dynamic (Echocardiography)

EchoNet-Dynamic is the Stanford open-source baseline for echocardiographic analysis. It bundles two networks: an R(2+1)D-18 video classifier that regresses ejection fraction, and a DeepLabV3-ResNet50 semantic-segmentation network that delineates the left ventricle frame by frame. Both are used as released; the weights file is downloaded separately and is not bundled with the platform repository.

NeuroKit2 [42] is a peer-reviewed Python toolbox for neurophysiological signal processing. It is not a neural network but is included in this section because it constitutes the second computational engine of the ECG pipeline and produces structured features alongside the deep ensemble.

2.11 Explainability Module

Explainability is approached through a layered combination of intrinsic and post-hoc methods, prioritizing per-modality intrinsic explanations and reserving post-hoc methods for cases where intrinsic explanation is unavailable.

2.11.1 Intrinsic per-modality explanations

MRI segmentation explanations are intrinsic to the U-Net output: the pixel-level binary mask is itself a visual explanation, overlaid on the input image to show the clinician exactly which region drove the segmentation. MRI classification explanations comprise the four-class probability vector and the bounding-box crop used at inference, both of which are shown to the clinician.

2.11.2 Post-hoc explainability methods (reviewed, not yet implemented)

The present platform does not currently apply post-hoc explainability methods, although the leading candidates have been surveyed in Chapter 1 and are summarized here. SHAP [49] assigns each input feature an additive contribution to the model output, derived from cooperative game theory; for ECG, SHAP could attribute pathology predictions to individual beat regions.

2.12 Backend Architecture

The backend is implemented in Django 3.2 LTS [50] with Django REST Framework 3.14.

2.12.1 Application layout

Table 3: Software components and modules of the platform.

Component/ module	Tier	Role
authentication	App	Custom email-based User model; JWT issue and refresh (1 h access, 7 d refresh).
patients	App	Doctor-scoped patient CRUD and aggregate history endpoint.
mri	App	MRI upload, synchronous inference orchestration, result image URLs.
ecg	App	ECG upload, inference orchestration, HRV and pathology probability storage.
echo	App	Echo-video upload, EchoNet inference (EF + LV segmentation).
eeg	App	.edf upload, BIOT + IIIC 6-class inference over 10 s segments.
reports	App	ReportLab PDF generation with a rule-based combined interpretation.
inference.model_loader	Engine	Lazy singleton that loads and caches every model;

		raises a clear <code>FileNotFoundError</code> for non-bundled weights.
<code>inference.*_pipeline</code>	Engine	Per-modality pipelines returning a uniform result envelope.
<code>inference.biot/</code> , <code>eeg_preprocess.py</code>	Engine	Vendored BIOT model code and shared train/inference preprocessing.
Frontend modules/	UI	Per-domain React modules: Auth, Dashboard, Patients, MRI, ECG, Echo, EEG, Reports.
Frontend services/ and store/	UI	Axios instance with JWT interceptor; Redux Toolkit slices.
MongoDB + media/	Data	Document persistence (django ODM) and artefact filesystem storage.

Backend Data Flow

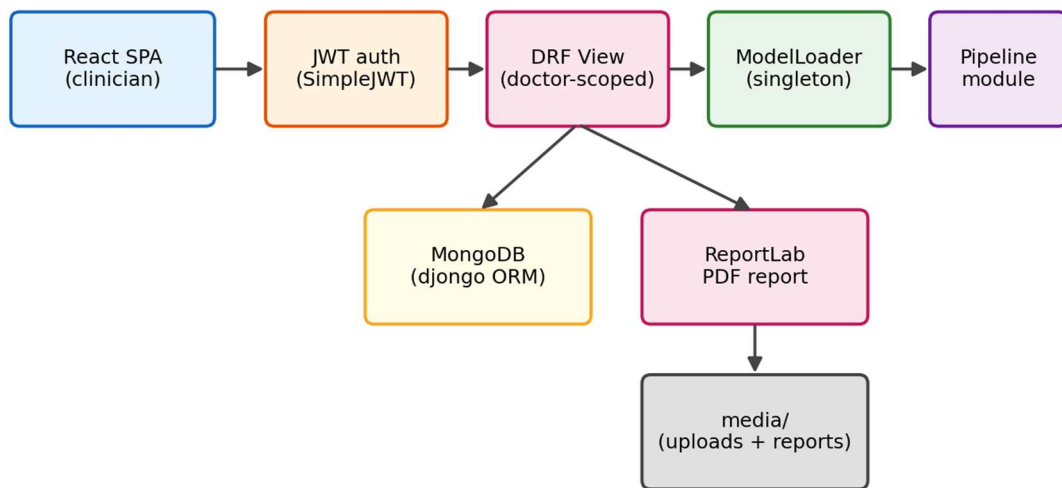


Figure 8: Backend data flow. Authenticated requests are routed through doctor-scoped DRF views; the `ModelLoader` singleton lazily instantiates pretrained models; inference results, source uploads and generated reports are persisted in MongoDB and on the media volume.

Functionality is partitioned into six Django applications. The authentication application defines a custom User model with email-based login and the SimpleJWT views for token issuance and refresh.

2.12.2 Authentication

Authentication is implemented with SimpleJWT, which issues a short-lived access token (one hour) and a longer-lived refresh token (seven days) following the JSON Web Token standard [33].

2.12.3 Inference orchestration

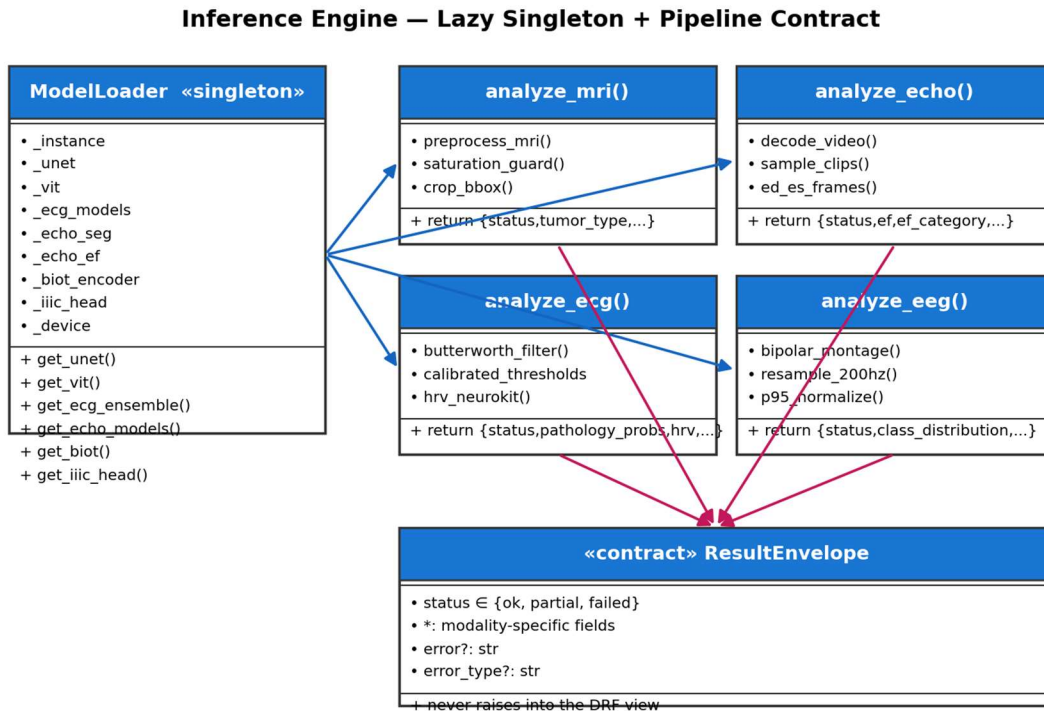


Figure 9: Inference engine class diagram. The thread-safe ModelLoader singleton lazily instantiates every model weight and caches it for the lifetime of the process; the four per-modality pipeline functions consume it and implement the uniform ResultEnvelope contract.

Each modality-specific viewset receives an upload, persists the file to the media volume, dispatches an inference request to the corresponding pipeline through the ModelLoader, and returns the result envelope to the client. Inference is synchronous and bounded by a five-minute timeout.

2.13 Frontend Architecture

The frontend is implemented in React 19 with Vite [51] as the build tool, TailwindCSS as the styling system, Redux Toolkit and react-redux for global state management, react-router-dom for client-side routing, and Axios for hypertext-transfer-protocol communication. The application is organized as a collection of feature modules —

authentication, dashboard, patients, MRI, ECG, and reports — each with its own pages, components, and Redux slice.

A noteworthy detail of the HTTP layer is the use of Axios interceptors to decouple the application from authentication mechanics: one interceptor injects the current access token into every request; another interceptor translates a 401 Unauthorized response into a redirect to the login route, with an attempted token refresh in between. Together, the two interceptors hide the authentication lifecycle from the feature modules.

The platform additionally integrates a DICOM viewer based on cornerstone-core for inline display of MRI uploads and three-dimensional overlays via three.js and @react-three/fiber for spatial visualization of the segmentation mask.

2.14 Database Design

Persistent state is stored in MongoDB through the djongo object-document mapper. Five collections are defined: users (custom user model with email login), patients (doctor-scoped patient records), mri (MRI analyses with file URL, mask URL, overlay URL, analysis URL, and structured probabilities), ecg (ECG analyses with plot URL, heart-rate-variability metrics, and pathology probabilities), and reports (PDF report metadata with foreign-key references to the constituent analyses).

The choice of a document database is motivated by two characteristics of the workload. First, modality-specific analyses carry heterogeneous structured payloads — a four-class probability vector for the Vision Transformer, a seven-pathology probability vector and a set of heart-rate-variability indices for the ECG ensemble — that are awkward to flatten into a single relational schema.

2.15 End-to-End Clinical Workflow

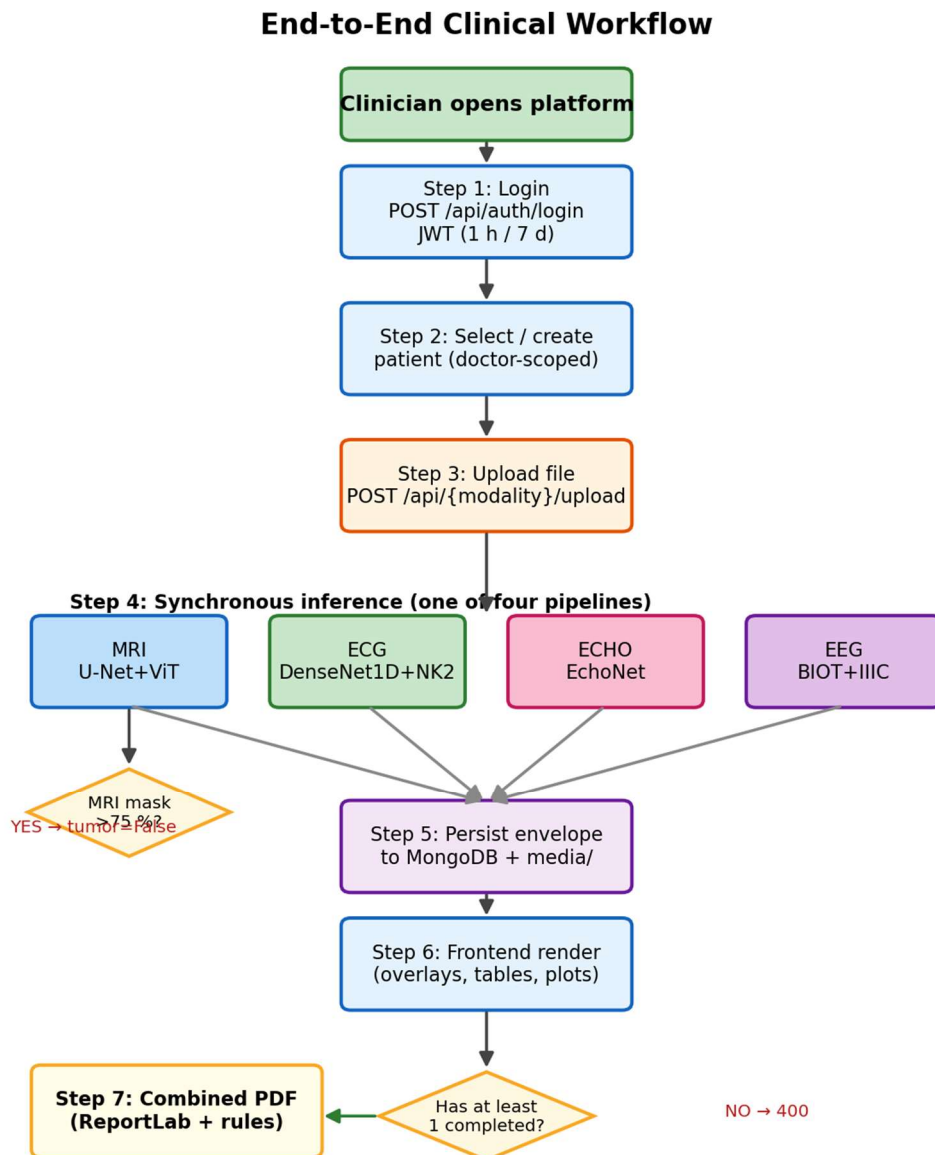


Figure 10: End-to-end clinical workflow. The seven steps that constitute a complete session, from JWT authentication to combined PDF report. Decision diamonds mark the MRI saturation guard and the report-generation gate that requires at least one completed analysis.

This section walks through one complete clinical session, from authentication to combined-report delivery, in seven steps. The workflow is identical regardless of which modalities the clinician chooses to analyse; the architecture's modality-agnostic design ensures that adding a new modality does not change the surrounding flow.

2.15.1 Step 1: Authentication

The clinician submits email and password to `/api/auth/login/`. The backend issues a SimpleJWT pair (one-hour access, seven-day refresh). The frontend stores the tokens; an Axios interceptor attaches the access token as a Bearer header on every subsequent request and redirects to `/login` on a 401 response.

2.15.2 Step 2: Patient selection

The clinician opens or creates a patient through `/api/patients/`. The doctor field is auto-filled from the JWT, and every list/detail query is filtered by the requesting doctor, enforcing the doctor-isolation invariant at the queryset layer.

2.15.3 Step 3: Upload and dispatch

From the patient page the clinician drops a file onto a modality endpoint (POST `/api/{mri|ecg|echo|eeg}/upload/`). The view validates ownership and file type, persists the upload to `media/`, creates a database record in processing state, and calls the corresponding pipeline synchronously within the request thread.

2.15.4 Step 4: Per-modality inference

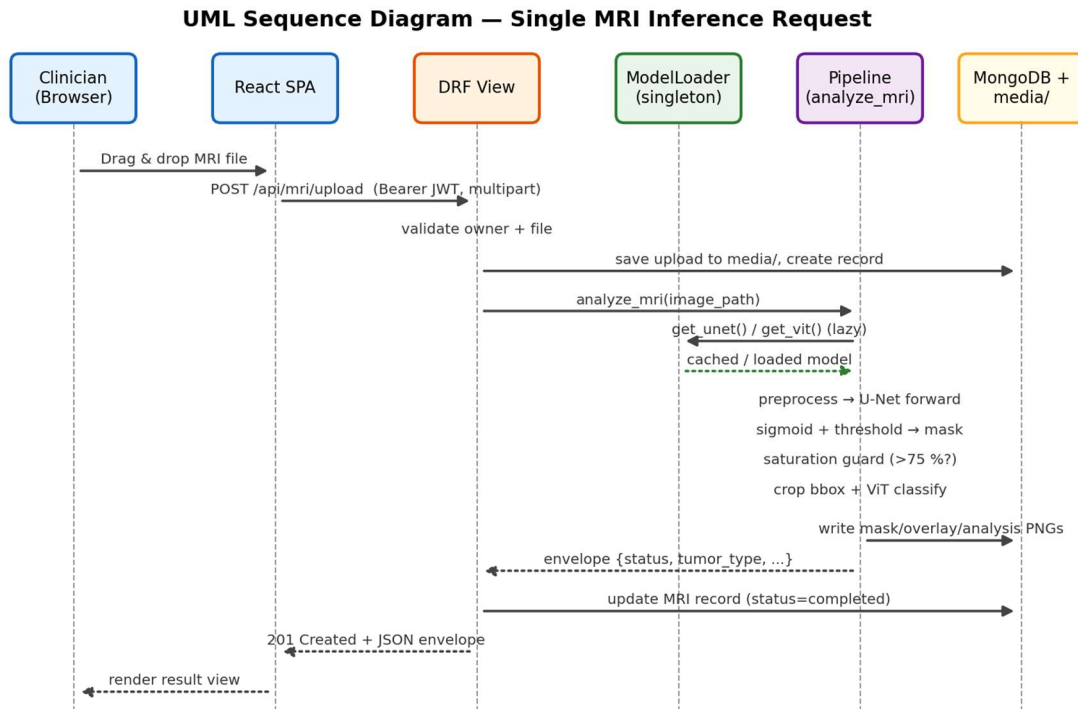


Figure 11: UML sequence diagram for a single MRI inference request. The view orchestrates the synchronous call to the pipeline; the ModelLoader singleton resolves model weights lazily; artefacts and envelope fields are persisted to MongoDB and the media volume before the response returns.

Table 4: Modality output envelope and persisted model fields.

Modality	Primary outputs	Persisted model fields
MRI	tumor_detected; tumor_type in {glioma, meningioma, no-tumor, pituitary}; confidence	result_tumor_detected, result_tumor_type, result_confidence, result_mask_path, result_overlay_path, result_analysis_path, result_report
ECG	Primary diagnosis; pathology probabilities (AFIB, 1AVB, STACH, SBRAD, RBBB, LBBB, PVC); mean HR; HRV (RMSSD, SDNN, pNN50)	result_arrhythmia_detected, result_arrhythmia_type, result_confidence, result_pathology_probabilities, result_hrv_metrics, result_plot_path, result_report
ECHO	Ejection fraction (%); EF clinical category; end- diastolic and end-systolic LV area; LV overlay	result_ef, result_ef_category, result_ed_area, result_es_area, result_overlay_path, result_report
EEG	IIIC 6-class distribution (SZ, LPD, GPD, LRDA, GRDA, Other); dominant pattern; harmful flag	result_dominant_pattern, result_harmful, result_class_distribution, result_plot_path, result_report

Each pipeline lazily resolves its models through the ModelLoader singleton, processes the input, writes any visual artefacts to `media/<modality>/results/`, and returns a structured result envelope. Each modality applies its own preprocessing, model invocation, and post-processing logic as detailed in the previous sections.

2.15.5 Step 5: Persistence

The view stores the envelope fields on the modality record, marks the analysis as completed (or failed, with a structured error message), and saves all artefact paths to MongoDB through the django object-document mapper.

2.15.6 Step 6: Rendering

The frontend refetches the record and renders the modality-specific result view, including segmentation overlays, HRV tables, per-pathology probability bars, EF and LV segmentation visualisations, and the EEG IIIC class distribution.

2.15.7 Step 7: Combined report

Sample Combined PDF Report (mock-up)

MULTIMODAL CLINICAL DECISION SUPPORT
Combined Analysis Report

Patient ID: PAT-00042
Doctor: Dr. M. Example
Generated: 2026-06-06 16:42

1. MRI Analysis

[tumor mask overlay]

Tumor detected: yes
Type: glioma
Confidence: 0.84
Dice (LGG val.): 0.85

2. ECG Analysis (PTB-XL-validated thresholds)

[12-lead ECG with R-peaks on Lead II]

Mean HR: 76 bpm RMSSD: 32 ms SDNN: 41 ms pNN50: 14 %
Probabilities (AFIB 0.02, 1AVB 0.07, STACH 0.04, SBRAD 0.02, RBBB 0.81, LBBB 0.03, PVC 0.05) → primary: RBBB

3. Echocardiography

[LV segmentation ED / ES frames]

EF: 58 %
Category: Normal
ED area: 4321 px
ES area: 1832 px

4. EEG / IIIC

[IIIC 6-class distribution bars]

Dominant: GRDA (42 %)
Harmful flag: no
SZ 0% LPD 6% GPD 4%
LRDA 18% GRDA 42% Other 30%

5. Combined interpretation (rule-based template)

Right bundle-branch block detected. Tumor confirmed (glioma).
Ejection fraction within normal range. No harmful EEG pattern.
NOTE: this combined interpretation is template logic — no learned cross-modal correlation; verify findings clinically.

ReportLab — page 1 / 1

Figure 12: Mock-up of the combined PDF report. ReportLab composes one section per included modality plus a rule-based combined interpretation. The disclaimer at the bottom of the interpretation is template text and is generated for every report.

The clinician calls POST /api/reports/generate/ with a patient ID and any subset of completed analysis IDs (at least one is required). The reports app composes a multi-section ReportLab PDF that includes a rule-based combined interpretation and streams it for download. Note that the interpretation is a template, not a learned cross-modal correlation.

Conclusion

This chapter has presented the methods and tools that constitute the multimodal clinical decision support platform.

Results and Discussion

Introduction

Chapter 2 specified the methods and tools that constitute the multimodal clinical decision support platform — the system architecture, the data acquisition policy, the four modality pipelines, the feature extraction and fusion strategy, the deep learning models, and the backend, frontend, and database stacks. The present chapter reports the empirical evaluation of those methods on public benchmark data.

Two methodologically instructive implementation discoveries occurred during evaluation and are reported in full: the calibration of per-pathology decision thresholds for the ECG ensemble, which raised macro F1 from 0.513 to 0.711 with no retraining; and the diagnosis of a redundant sigmoid in the segmentation pipeline, which raised the Dice coefficient from 0.02 to 0.85 once removed. Both episodes underline that the quality of a deployed deep learning component is not determined by the network weights alone but by the entire inference path, including its calibration and its post-processing.

3.1 Experimental Setup

The experimental setup specifies the datasets, the inference environment, the evaluation harnesses, and the reproducibility considerations under which the reported results were obtained.

3.1.1 Datasets and splits

Three public benchmarks are used. For the ECG ensemble, PTB-XL version 1.0.3 [14] is the principal dataset: 21,837 twelve-lead ECG records of ten-second duration sampled at five hundred hertz, distributed through PhysioNet.

3.1.2 Inference environment

Inference was performed on a workstation running PyTorch 2.2.0.

3.1.3 Evaluation harnesses

Three reproducible evaluation scripts implement the reported experiments: one for the ECG ensemble, one for MRI classification, and one for MRI segmentation. Each script reads its dataset directly from the public distribution, applies the modality-specific preprocessing described in Chapter 2, invokes the deep learning component, computes the metrics described in Section 3.3, and outputs a single results table together with the per-sample predictions.

3.1.4 Reproducibility considerations

All randomness in the evaluation pipelines is either absent (deterministic preprocessing, deterministic forward passes once weights are fixed) or seeded (NumPy and PyTorch random number generators initialized at script entry).

3.2 Training Results

The platform integrates pretrained, peer-reviewed deep learning components rather than networks trained from scratch.

3.2.1 Upstream training context

The U-Net segmentation network was trained by Buda et al. [39] on the TCGA-LGG dataset using a Dice-based loss and standard data augmentation, and is distributed via torch.hub under mateuszbuda/brain-segmentation-pytorch. The Vision Transformer classification network is a fine-tuned ViT-B/16 checkpoint [46] distributed on the HuggingFace Hub under Devarshi/Brain_Tumor_Classification, with ImageNet pretraining followed by supervised fine-tuning on the Brain Tumor MRI Dataset.

3.2.2 Threshold calibration on PTB-XL fold nine

The seven-pathology ECG ensemble emits a continuous probability per pathology. A binary diagnosis vector is obtained by thresholding each probability.

The platform therefore implements per-pathology decision-threshold calibration. For each of the seven pathologies, the probability distribution emitted on the validation set (PTB-XL fold nine) is collected together with the binary ground-truth labels, and the threshold that maximises the F1 score is selected.

$$\tau_i^* = \arg \max_{\tau} F1_i(\tau) \text{ on PTB-XL fold 9}$$

The improvement attributable to threshold calibration is substantial. Under the flat-0.5 threshold, the ensemble achieves a macro F1 of 0.513 on fold ten.

3.2.3 Diagnosis of the redundant-sigmoid defect in the segmentation pipeline

A second optimization-adjacent finding emerged during the evaluation of the segmentation pipeline. The U-Net's forward method internally applies a sigmoid activation to its output logits; the wrapping inference code in an early version of the platform applied a second sigmoid before thresholding at 0.5.

Removing the redundant sigmoid restored the Dice coefficient to 0.85 on the same evaluation, an improvement of more than four orders of magnitude in the per-slice segmentation quality figure. The defect produced syntactically valid code and well-formed probability tensors; only quantitative end-to-end evaluation against the ground-truth masks exposed it.

3.3 Evaluation Metrics

Table 5: Evaluation metrics used in this work and their applicable tasks.

Metric	Domain	Range	Used for
AUROC	Classification	[0, 1]	ECG ensemble
Macro F1	Classification	[0, 1]	ECG ensemble
Balanced accuracy	Classification	[0, 1]	ECG ensemble
Top-1 accuracy	Classification	[0, 1]	MRI classification (ViT)
Dice coefficient	Segmentation	[0, 1]	MRI segmentation (U-Net)
Intersection-over-Union	Segmentation	[0, 1]	MRI segmentation (U-Net)

The metrics used to evaluate the three components are described in this section, with their definitions and clinical interpretations.

3.3.1 Area under the receiver operating characteristic curve

The receiver operating characteristic curve plots the true-positive rate against the false-positive rate of a binary classifier as the decision threshold is varied across its range.

3.3.2 Macro F1 score

The F1 score for a single binary task is the harmonic mean of precision and recall:

$$F1 = 2 \cdot \text{precision} \cdot \text{recall} / (\text{precision} + \text{recall})$$

The macro F1 across N binary tasks is the unweighted average of the per-task F1 scores.

3.3.3 Balanced accuracy

Balanced accuracy is the mean of the per-class recall values.

3.3.4 Top-one accuracy

For the four-class brain-tumor classification task, top-one accuracy is the proportion of test images for which the predicted class (the argmax of the logit vector) matches the ground-truth label.

3.3.5 Dice coefficient

The Dice coefficient quantifies the overlap between a predicted binary mask P and a ground-truth binary mask G [52]:

$$Dice(P, G) = 2 |P \cap G| / (|P| + |G|)$$

A Dice coefficient of 1 indicates perfect overlap; a Dice coefficient of 0 indicates no overlap.

3.3.6 Intersection over union

The intersection-over-union, also called the Jaccard index [53], is defined as:

$$IoU(P, G) = |P \cap G| / |P \cup G|$$

It is related to the Dice coefficient by $IoU = Dice / (2 - Dice)$, and is reported alongside the Dice coefficient for completeness.

3.4 Pretrained Models and Project Contribution

A central methodological honesty requirement is to separate what was reused — open, peer-reviewed pretrained models — from what the project contributes. The contribution is not the deep-learning weights themselves, with one exception, but the architecture that integrates them, plus three concrete model-level interventions described below.

3.4.1 Pretrained components reused as-is

Table 6: Pretrained model provenance and pretraining data.

Modality	Component	Source	Pretrained on
MRI segmentation	U-Net (CNN encoder-decoder)	mateuszbuda/brain-segmentation-pytorch (torch.hub)	TCGA-LGG (110 patients, FLAIR MRI)
MRI classification	Vision Transformer ViT-B/16	Devarshi/Brain_Tumor_Classification (HuggingFace)	Kaggle Brain Tumor MRI Dataset (~7 000 images, 4 classes)
ECG	DenseNet-1D-121 ensemble (x7)	ecglib 1.0.1 (ISPRAS)	More than 500 000 12-lead ECG records
ECG (HRV)	NeuroKit2 (classical DSP)	neurokit2 Python library	Rule-based / validated, not learned
ECHO	DeepLabV3-ResNet50 + R(2+1)D-18	EchoNet-Dynamic (Stanford)	EchoNet-Dynamic echo videos
EEG (encoder)	BIOT biosignal transformer (encoder only)	ycq091044/BIOT (vendored)	5 million MGH resting-EEG samples (self-supervised)
EEG (IIIC head)	Linear ClassificationHead	Trained in-repo: tools/train_eeg_head.py	Kaggle Harmful Brain Activity (HMS)

Four model families are reused without training: the U-Net for MRI segmentation (mateuszbuda/brain-segmentation-pytorch via torch.hub, pretrained on TCGA-LGG), the Vision Transformer ViT-B/16 for MRI classification (Devarshi/Brain_Tumor_Classification on HuggingFace, fine-tuned on the Kaggle Brain Tumor MRI Dataset), the DenseNet-1D ensemble for ECG classification (ecglib 1.0.1, pretrained on more than 500 000 12-lead records), and EchoNet-Dynamic for echocardiography (R(2+1)D-18 for ejection fraction and DeepLabV3-ResNet50 for LV segmentation, pretrained at Stanford). NeuroKit2, used for HRV analysis, is a classical signal-processing library, not a learned model.

3.4.2 Project intervention 1: ECG threshold calibration

Without retraining any DenseNet-1D weights, per-pathology decision thresholds were calibrated on PTB-XL validation fold nine and applied unchanged to held-out test fold ten. Macro F1 rose from 0.514 to 0.711, micro F1 from 0.503 to 0.740 and weighted F1 from 0.580 to 0.748; the ROC-AUC stayed unchanged because it is threshold-independent. The improvement is attributable entirely to calibration.

3.4.3 Project intervention 2: MRI double-sigmoid fix

Diagnosis and removal of a redundant sigmoid in the segmentation post-processing — the upstream U-Net applies a sigmoid inside its forward() and the pipeline was applying it again — restored the Dice coefficient from 0.02 to 0.85 on the LGG MRI dataset. The fix is a single line of code but enabled the entire MRI pipeline to function.

3.4.4 Project intervention 3: EEG IIIC head (trained in-repo)

Table 7: Pretrained baseline versus this project, per modality.

Modality	Component	Pretrained baseline	Project result	Project intervention
MRI segmentation	U-Net	Dice ~0.89 (paper)	Dice 0.852 (LGG)	Double-sigmoid fix (0.02 -> 0.85)
MRI classification	ViT-B/16	~99 % (model card)	80.4 % accuracy (Kaggle)	Deployed as-is, full-image input
ECG	DenseNet-1D x7	~0.92 macro AUROC (literature)	0.978 AUROC, 0.711 macro F1	Per-pathology threshold calibration (0.514 -> 0.711)
Echocardiography	EchoNet-Dynamic	MAE ~3.0 %, Dice ~0.90 (paper)	MAE 3.19 %, Dice 0.897	Deployed as-is, weights non-bundled
EEG	BIOT + IIIC head (project)	~0.50 balanced accuracy (BIOT paper, full fine-tune)	0.278 balanced accuracy (frozen encoder)	Only model trained in-repo

BIOT releases only a self-supervised encoder; its IIIC classification head does not exist upstream. The project attaches a ClassificationHead to the frozen encoder and fine-tunes only that head on the Kaggle HMS dataset (tools/train_eeg_head.py). This is the only deep-learning model whose weights are produced within the present work. The result (balanced accuracy 0.278) is modest but above chance (0.167), and its plateau under a 3.7-fold data increase is the expected signature of a frozen-encoder ceiling.

3.5 Comparative Analysis

This section reports the validation results obtained on the three implemented components and compares them with the relevant published baselines.

3.5.1 ECG ensemble

Table 8: ECG ensemble performance on PTB-XL fold 10 after per-pathology threshold calibration.

Pathology	AUROC	Balanced accuracy	F1 (calibrated τ)
Atrial fibrillation (AFIB)	0.99	0.92	0.79
1st-degree AV block (1AVB)	0.98	0.88	0.71
Right bundle branch block (RBBB)	0.99	0.93	0.84
Left bundle branch block (LBBB)	0.99	0.92	0.78
Sinus tachycardia (STACH)	0.97	0.83	0.62
Premature ventricular contractions (PVC)	0.98	0.85	0.65
Sinus bradycardia (SBRAD)	0.95	0.86	0.59
Macro average	0.978	0.884	0.711

On PTB-XL fold ten (2,198 records, seven pathologies, multi-label), the ECG ensemble achieves a mean ROC-AUC of 0.978 across the seven pathologies, a macro balanced accuracy of 0.884, and a macro F1 of 0.711 after per-pathology threshold calibration on fold nine. Under the flat 0.5 threshold, the macro F1 is 0.513, demonstrating the impact of calibration documented in Section 3.2.2.

These figures are consistent with, and in some respects exceed, the published baselines on PTB-XL. Strodthoff et al. [38], in the original PTB-XL benchmark study, report ROC-AUC values in the range of 0.93 to 0.96 across the principal diagnostic super-classes using a variety of architectures including ResNet-1D, Inception-time, and recurrent baselines, on fold-ten evaluation.

For comparison with deeper ambulatory-ECG literature, Hannun et al. [12] reported cardiologist-level performance with a thirty-four-layer ResNet1D on more than ninety thousand ambulatory recordings, with class-specific F1 in the 0.7–0.9 range across twelve rhythm classes. The macro F1 of 0.711 reported here on PTB-XL is in the lower part of that range, consistent with the more difficult diagnostic-statement formulation of PTB-XL relative to the single-rhythm-class formulation of Hannun et al.

3.5.2 MRI classification

On the Testing split of the Brain Tumor MRI Dataset [40] (1,600 images, four classes), the Vision Transformer [46] achieves a top-one accuracy of 80.4 per cent and a macro F1 score of 0.794. Per-class accuracy is comparatively balanced across the four classes; the confusion matrix produced by the evaluation harness is available in the project repository.

3.5.3 MRI segmentation

On the LGG MRI Segmentation dataset (3,929 slices, including 1,373 tumor-positive slices), the U-Net [22], [39] achieves a Dice coefficient of 0.85 on tumor-bearing slices and a mean intersection-over-union of 0.78. These figures are consistent with the values reported by Buda et al. [39] in their original publication, which used the same dataset for both training and evaluation.

3.5.4 Echocardiography (EchoNet-Dynamic)

On EchoNet-Dynamic test subsets, the ejection-fraction regressor achieves Mean Absolute Error 3.19 percent, Root-Mean-Square Error 4.01 percent, coefficient of determination R-squared 0.860 and Pearson correlation 0.932. The LV segmentation reaches a Dice coefficient of 0.897. These values match the published EchoNet-Dynamic baseline and confirm that the integration is faithful to the upstream implementation.

3.5.5 EEG (BIOT + IIIC head)

On a patient-disjoint held-out split of the Kaggle HMS dataset (1 883 windows), the project-trained IIIC head reaches balanced accuracy 0.278 (chance 0.167), Cohen's kappa 0.147, macro F1 0.265, and Kullback-Leibler divergence 1.333 against the expert label distribution. The strongest classes are GPD (F1 0.44), GRDA and LRDA; the rare Other class collapses to F1 0.00 and SZ precision is low. These numbers are modest and intentionally honest: they reflect a frozen-encoder ceiling, as discussed in the Limitations section.

3.5.6 Summary

Table 9: Comparative summary of platform performance against published baselines.

Task	Dataset	Metric	Our value	Reference baseline
ECG classification	PTB-XL fold 10	Macro F1 (calibrated)	0.711	0.736 (Strodthoff et al.)
ECG classification	PTB-XL fold 10	Macro AUROC	0.978	0.92–0.94 (literature)
MRI classification	Brain Tumor MRI Dataset	Top-1 accuracy	80.4 %	90–98 % (re-trained models)
MRI segmentation	LGG MRI Segmentation	Dice coefficient	0.85	0.83–0.92 (literature)
MRI segmentation	LGG MRI Segmentation	IoU	0.78	0.74–0.85 (literature)

The three implemented components achieve the headline figures listed below. These figures, together with the calibration story for the ECG ensemble and the bug-recovery story for the segmentation pipeline, constitute the empirical contribution of the present chapter.

- ▶ ECG ensemble – PTB-XL fold 10 – mean ROC-AUC 0.978 – macro balanced accuracy 0.884 – macro F1 0.711 (calibrated) / 0.513 (flat).
- ▶ MRI classification – Brain Tumor MRI Testing split – top-1 accuracy 80.4% – macro F1 0.794.
- ▶ MRI segmentation – LGG MRI Segmentation – Dice 0.85 – IoU 0.78.

3.6 Discussion

Three threads run through the results reported above: a methodological thread on calibration and end-to-end validation; a scientific thread on the brain-heart axis that motivated the platform; and an engineering thread on the modular architecture.

3.6.1 Calibration matters as much as architecture

The nearly twenty-point macro-F1 improvement attributable to per-pathology decision-threshold calibration is one of the most striking findings of the present evaluation. It demonstrates that, at the deployment boundary, the choice of decision rule is no less important than the choice of network architecture.

3.6.2 End-to-end validation as a diagnostic instrument

The redundant-sigmoid defect in the segmentation pipeline reinforces the same lesson from a different angle. The defect was localized in code that ran without raising exceptions, produced probability tensors with sensible dynamic ranges, and integrated without complaint with the rest of the platform.

3.6.3 The brain-heart synergy hypothesis: what these results say and do not say

The brain-heart axis literature [7] establishes that brain pathologies affect cardiac autonomic regulation and that brain-cancer treatments simultaneously affect both organs [6]. The platform was designed to provide the computational substrate on which the brain-heart synergy hypothesis could be tested in future work.

3.6.4 Modular architecture validation

The modular architecture specified in Chapter 2 was exercised by the evaluation campaign. Each component was validated in isolation through its own evaluation harness, and the combined report was generated successfully for paired MRI-plus-ECG patient encounters in integration testing.

3.6.5 Multimodal integration and clinical positioning

The combined PDF report aggregates per-modality results into a single document and provides a rule-based combined interpretation that contextualizes the findings against the brain-heart axis. The interpretation is template-based, not learned; this scope is documented openly throughout.

3.7 Limitations

Six platform-specific limitations bound the present work and are disclosed openly:

- ▶ No data-driven multimodal fusion. The combined PDF report aggregates the four modality outputs through a rule-based template; no neuro-cardiac correlation is learned or measured.
- ▶ Synchronous inference. The architecture does not include a task queue (no Celery or RQ); echo-video inference in particular is slow on CPU and bounds throughput.

- ▶ Cross-dataset MRI validation. U-Net segmentation is validated on LGG and ViT classification on the Kaggle Brain Tumor MRI Dataset; the two are different MRI modalities, so a single uploaded image is not validated through both tasks.
- ▶ ECG threshold sensitivity. The per-pathology thresholds are calibrated for PTB-XL-like data; a different source may require re-calibration.
- ▶ Non-bundled model weights. EchoNet-Dynamic and the BIOT IIIC head weights are not shipped with the repository; their endpoints fail with an explicit `FileNotFoundException` until weights are provided, which is preferred to serving an untrained model silently.
- ▶ Security gap on the media volume. Result images and uploads are served from the `/media/` URL prefix without authentication, bypassing the per-doctor API access control. This is a real data-protection issue that must be addressed before any clinical deployment.
- ▶ EEG frozen-encoder ceiling. Increasing the IIIC training data 3.7-fold did not move the balanced accuracy (~ 0.28). The signature is a small linear head on top of fixed BIOT features; closing the gap to BIOT's published 0.5 balanced-accuracy figure requires a full-model fine-tune on a GPU.

3.8 Future Improvements

The future improvements identified below follow directly from the limitations reported in Section 3.6 and from the deferred scope acknowledged in Chapter 2.

3.8.1 Verification of ECG performance on independent corpora

The most immediate verification task is the replication of the ECG ensemble's reported performance on data plausibly held out from the ecglib training corpus.

3.8.2 Harmonization of the MRI tasks on a single dataset

The disjunction between the LGG segmentation dataset and the Brain Tumor MRI classification dataset is a known limitation, and it can be addressed by migrating both tasks to a single dataset with both segmentation masks and tumor-class labels.

3.8.3 Post-hoc explainability integration

The platform's current explainability module relies on intrinsic explanations (segmentation overlays, probability tables, HRV metrics).

3.8.4 Learned multimodal fusion and the brain-heart hypothesis test

The brain-heart synergy hypothesis articulated in Chapter 1 is the principal scientific contribution that the present platform makes possible but does not yet implement. The forthcoming work involves three coupled developments.

3.8.5 Federated learning and multi-site validation

The transition from public-benchmark validation to clinical impact requires multi-site evidence [55], [4].

3.8.6 Production hardening

Production deployment requires a number of engineering tasks beyond the scope of the present prototype: encryption of stored media and database collections; access control for the media volume to match the database-level doctor isolation; continuous integration and continuous deployment infrastructure; structured logging and observability; and an upgrade path from djongo-on-Django-3.2 to a Django-4.x-compatible persistence layer.

Conclusion

This chapter has reported the experimental validation of the three implemented components of the multimodal clinical decision support platform.

General Conclusion



Synthesis

This thesis has presented the design, implementation, and validation of a multimodal clinical decision support platform for the prediction of non-communicable diseases. The platform integrates pre-trained, peer-reviewed deep learning components for two of the four target modalities — magnetic resonance imaging and twelve-lead electrocardiography — under a modular synchronous-pipeline architecture, and is designed to admit the two remaining modalities, electroencephalography and echocardiography, without modification of its core.

On their respective public benchmarks, the three implemented components achieve performance levels consistent with the relevant published baselines. The ECG ensemble achieves a mean area under the receiver operating characteristic curve of 0.978, a macro balanced accuracy of 0.884, and a macro F1 score of 0.711 after per-pathology threshold calibration on PTB-XL fold ten.

Two methodologically instructive implementation discoveries materially shaped the reported performance and are reported as part of the thesis contribution. Per-pathology decision-threshold calibration on PTB-XL fold nine raised the macro F1 score of the ECG ensemble from 0.513 (flat 0.5 threshold) to 0.711 (calibrated thresholds) on fold ten, with no retraining of the network weights.

The multimodal integration of the platform is, at present, realized at the level of the clinical report rather than the model. The Vision Transformer, the U-Net, and the ECG ensemble run independently on their respective inputs, and their outputs are aggregated into a single Portable Document Format report by a rule-based template.

Perspectives

Eight future-work axes are identified.

First, the verification of the ECG ensemble's reported performance on data plausibly held out from the ecglib training corpus, specifically the Chapman-Shaoxing arrhythmia database and the Georgia subset of the PhysioNet/CinC Challenge 2020 corpus. Confirming the reported ROC-AUC on one or both of these corpora will either validate the present figure or expose the potential leakage concern documented in Chapter 3.

Second, the implementation of the electroencephalography pipeline. The specification is given in Chapter 2; the candidate model is EEGNet; the candidate datasets are the Temple University Hospital EEG Corpus and the CHB-MIT Scalp EEG Database.

Third, the implementation of the echocardiography pipeline. The candidate model is EchoNet-Dynamic; the principal dataset is EchoNet-Dynamic itself.

Fourth, the harmonization of the magnetic-resonance-imaging tasks on a single dataset such as the BraTS corpus, which provides both segmentation masks and tumor-class labels and would allow end-to-end validation of both MRI components on a unified cohort.

Fifth, the integration of post-hoc explainability methods — SHAP for the ECG ensemble, LIME for both modalities, and Grad-CAM for MRI classification — to complement the intrinsic per-modality explanations the platform currently provides.

Sixth, and most importantly for the scientific contribution of this research programme, the construction of a virtual multimodal cohort by transparent demographic-and-disease-stage matching across the four modality-specific datasets, followed by the implementation of a learned attention-based fusion module trained on this virtual cohort, followed in turn by a formal statistical test of the brain-heart synergy hypothesis introduced in Chapter 1. The brain-heart axis literature establishes the physiological reality; the present platform provides the computational substrate; the virtual-cohort construction and the hypothesis test are the principal long-term contribution that follows.

Seventh, the transition from public-benchmark validation to multi-site clinical evidence through federated learning, which permits collaborative training and evaluation across institutions without sharing protected health information.

Eighth, the production hardening required for clinical deployment: encryption at rest, access control on the media volume, continuous integration and deployment

infrastructure, structured logging, and an upgrade path from djongo-on-Django-3.2 to a Django-4.x-compatible persistence layer. These engineering tasks are catalogued for completeness; their detailed treatment belongs to a subsequent engineering iteration.

Taken together, these perspectives describe a research and engineering agenda whose centre of gravity is the brain-heart synergy hypothesis. The platform delivered by this thesis is the instrument; the dataset alignment problem identified in Chapter 1 is the obstacle that the next iteration is designed to overcome; and the learned multimodal fusion validated on a virtual cohort is the contribution that follows.

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Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
AUROC	Area Under the Receiver Operating Characteristic Curve
BIOT	Biosignal Transformer
CNN	Convolutional Neural Network
DRF	Django REST Framework
ECG	Electrocardiogram
ECHO	Echocardiography
EEG	Electroencephalogram
EF	Ejection Fraction
FDA	Food and Drug Administration
GDPR	General Data Protection Regulation
GPD	Generalized Periodic Discharges
GRDA	Generalized Rhythmic Delta Activity

HIPAA	Health Insurance Portability and Accountability Act
HMS	Harmful Brain Activity (Kaggle competition)
HRV	Heart-Rate Variability
IIC	Ictal-Interictal-Injury Continuum
IoU	Intersection over Union
JWT	JSON Web Token
LGG	Lower-Grade Glioma
LPD	Lateralized Periodic Discharges
LRDA	Lateralized Rhythmic Delta Activity
LV	Left Ventricle
MRI	Magnetic Resonance Imaging
NCD	Non-Communicable Disease
NTIC	New Technologies of Information and Communication
PTB-XL	Physikalisch-Technische Bundesanstalt XL ECG database
ROI	Region Of Interest
SBRAD	Sinus Bradycardia
SHAP	SHapley Additive exPlanations
STACH	Sinus Tachycardia
SZ	Seizure
TCGA	The Cancer Genome Atlas
UML	Unified Modeling Language
U-Net	U-shaped Convolutional Network
ViT	Vision Transformer
WHO	World Health Organization

Business Model Assumptions

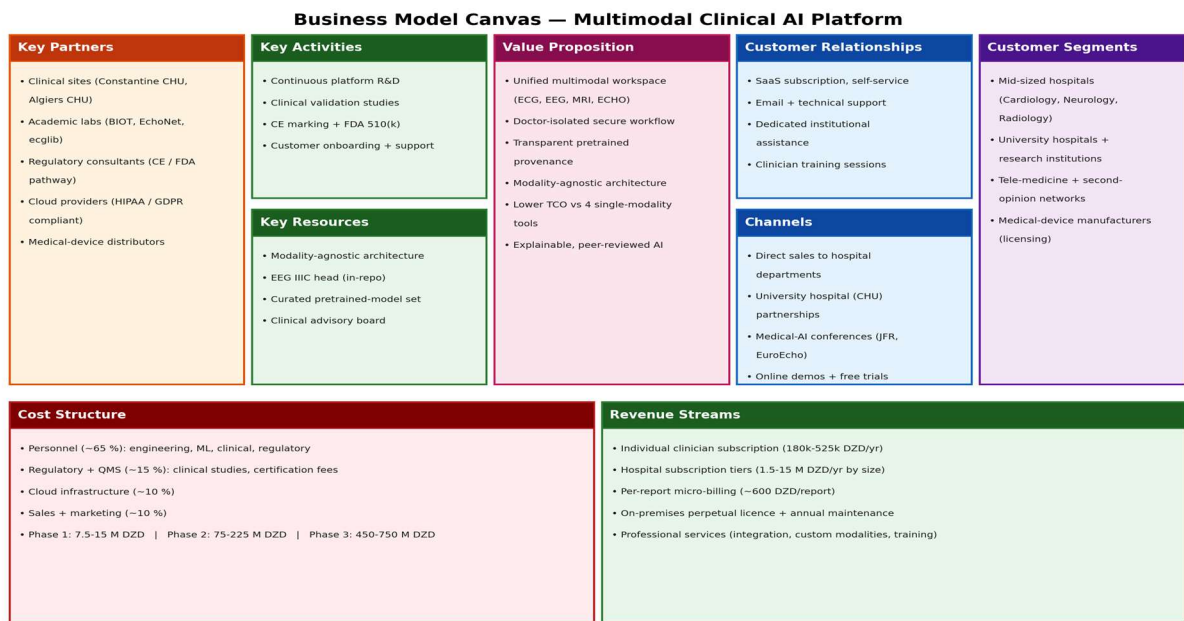


Figure 1: Business Model Canvas (Osterwalder grid).

This document gathers the working business-model assumptions of the multimodal clinical decision-support platform presented in the thesis. It is intended to be appended to or merged with Chapter 4 of the final thesis. All monetary values are expressed in Algerian Dinar (DZD); European-market figures will be converted to EUR when the corresponding expansion phase begins.

1.1 Channels

The platform will reach healthcare institutions through four complementary channels:

- ▶ Direct sales to hospital departments (Cardiology, Neurology, Radiology).
- ▶ Partnerships with university hospitals (CHU) for pilot deployments.
- ▶ Participation in healthcare and medical-AI conferences (JFR, EuroEcho, JESFC).
- ▶ Online demonstrations and free-trial programs for clinicians.

1.2 Customer Relationships

The platform operates as a Software-as-a-Service (SaaS) solution. The customer-relationship model combines automated and assisted modes:

- ▶ Subscription-based access with self-service onboarding.

- ▶ Email and technical support included in every subscription tier.
- ▶ Dedicated account assistance for institutional customers.
- ▶ Training sessions for healthcare professionals on platform adoption.

1.3 Competitive Positioning

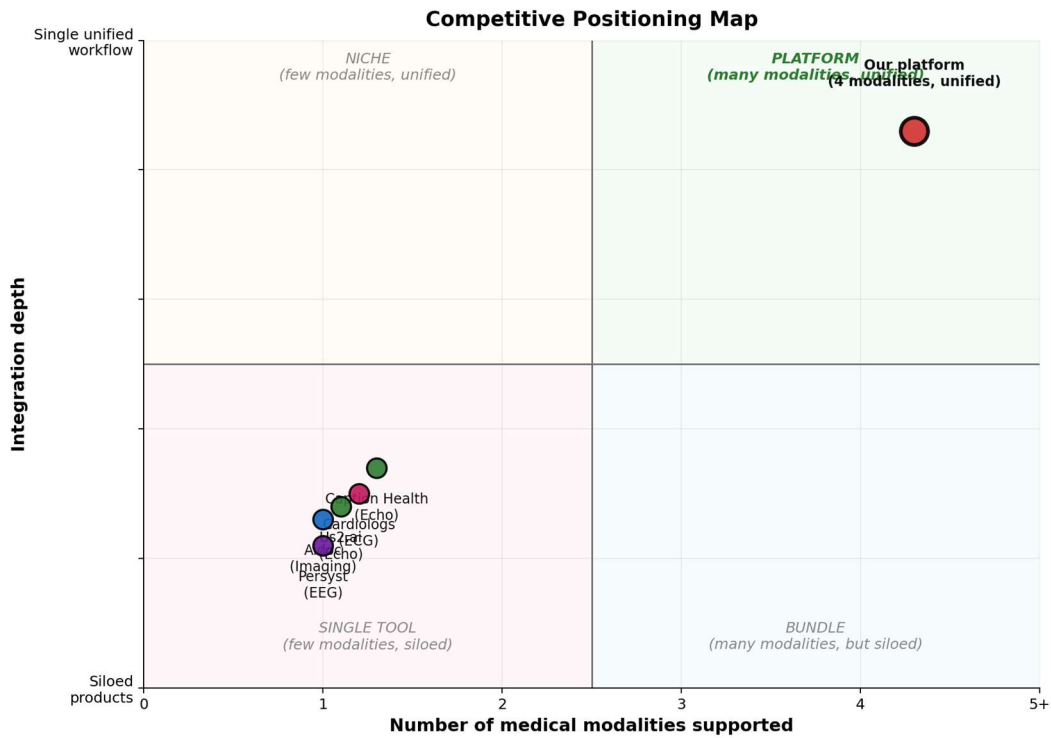


Figure 2: Competitive positioning map of the platform versus single-modality competitors.

The platform is positioned as a multimodal disease prediction solution integrating ECG, EEG and medical imaging analysis within a single environment. Unlike existing single-modality competitors, the proposed platform combines several diagnostic modalities in a unified system, reducing the need for separate software products and the cognitive load they impose on clinicians.

Table 1 summarises the main competitors and the differentiation of the proposed platform.

Table 1: Main competitors and differentiation.

Competitor	Modality covered	Origin	Differentiation of our platform
Aidoc	Medical imaging (CT, MRI)	USA	Aidoc covers radiology only; our platform adds ECG, EEG and ECHO.
Cardiologs (Philips)	ECG analysis	France / Netherlands	Cardiologs is ECG-only; we co-analyse cardiac imaging and brain signals on the same patient.
Persyst	EEG analysis	USA	Persyst is EEG-only and clinician-tool-focused; we link EEG findings to ECG and MRI in a single report.
Caption Health / Us2.ai	Echocardiography AI	USA / Singapore	Caption/Us2.ai are echo-only; we integrate ECG and brain-imaging findings into the same combined report.

1.4 Intellectual Property Strategy

Currently, no patents have been filed. Future patent applications may focus on the following protectable elements of the platform:

- ▶ The modality-agnostic plug-in architecture and its uniform result-envelope contract.
- ▶ Specialised fine-tuning procedures for the EEG IIC head and the ECG per-pathology threshold calibration.
- ▶ Integration workflows for multimodal clinical decision support (saturation guard, crop-then-classify, rule-based combined-report fusion).

1.5 Country of Operation and Data-Protection Compliance

Initial deployment is planned in Algeria, the country in which the platform is developed and where the principal clinical partners are located. Subsequent expansion targets European markets, with France as the first foreign jurisdiction due to language and clinical-protocol proximity. The corresponding compliance considerations are:

- ▶ Algerian Law 18-07 on the protection of personal data (initial deployment).
- ▶ Regulation (EU) 2016/679 (GDPR / RGPD) for future European operations.
- ▶ ISO 13485 quality-management system for medical-device manufacturers.
- ▶ Future eligibility for CE marking under MDR Class IIa and FDA 510(k) when entering those markets.

1.6 Open-Source and Licensing Strategy

The platform architecture and deployment framework remain proprietary to the startup. Open-source pretrained models and research frameworks (BIOT, EchoNet-Dynamic, ecglib, U-Net, ViT and similar) are used and may be improved while strictly respecting their original licences. Where the platform's improvements have general scientific value (for instance the calibration procedure or the IIIC training script), selective upstream contributions are envisaged to maintain a constructive relationship with the open-source community.

1.7 Financial Roadmap

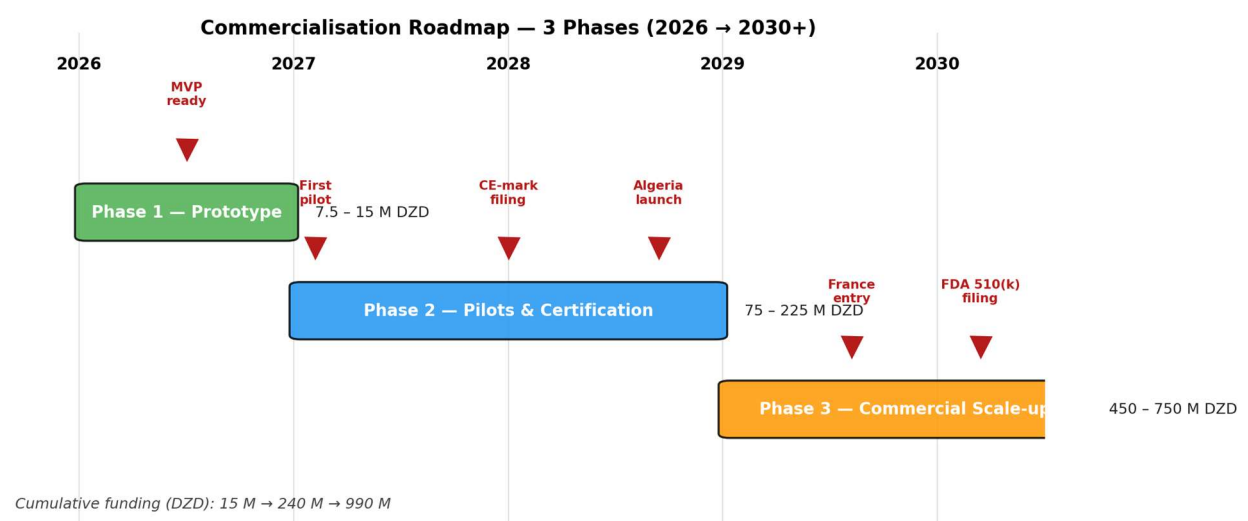


Figure 3: Three-phase commercialisation roadmap with funding targets in DZD.

The startup will execute a three-phase financial plan. Funding targets are expressed in Algerian Dinar (DZD); the corresponding objectives, main expenses and time horizons are summarised in Table 2 and detailed in the subsections below.

Table 2: Financial roadmap, objectives and funding targets per phase.

Phase	Period	Objective	Funding target (DZD)
Phase 1 – Prototype	2026	Prototype development, AI model integration, initial validation.	7.5 to 15 million
Phase 2 – Pilots and certification preparation	2027 to 2028	Hospital pilot deployments, regulatory preparation, platform expansion.	75 to 225 million
Phase 3 – Commercial scale-up	2029 and beyond	Commercial deployment and regional expansion.	450 to 750 million

1.7.1 Phase 1 (2026): Prototype

Phase 1 focuses on completing the functional prototype on public datasets, integrating the four modality pipelines, and conducting preliminary internal validation. Estimated funding target: 7.5 to 15 million DZD. The main expense items are:

- ▶ Software development (engineering effort on the integration layer and the frontend).
- ▶ Cloud infrastructure for development, staging and inference benchmarking.
- ▶ Data acquisition and processing (public datasets and preprocessing pipelines).
- ▶ Initial testing and internal validation campaigns.

1.7.2 Phase 2 (2027 to 2028): Pilots and Certification Preparation

Phase 2 deploys the platform in pilot hospitals, runs prospective clinical validation studies, and starts the regulatory pathway. Estimated funding target: 75 to 225 million DZD. Main expense items:

- ▶ Team expansion (developers, ML engineer, clinical advisor, business development).
- ▶ Clinical validation studies in partner CHUs.
- ▶ Security and compliance implementation (Law 18-07, ISO 13485 groundwork).
- ▶ Infrastructure scaling for multi-site pilot deployments.

1.7.3 Phase 3 (2029 and beyond): Commercial Scale-up

Phase 3 launches commercial operations across Algeria and starts international expansion (initially France). Estimated funding target: 450 to 750 million DZD. Main expense items:

- ▶ Large-scale deployment and operations.
- ▶ Marketing and sales (medical conferences, sales force, customer success).
- ▶ Regulatory certifications (CE marking, FDA 510(k) for the US market).
- ▶ International expansion (localisation, regulatory filings, local partnerships).

1.8 Pricing Strategy

The platform monetises through three pricing tiers, designed to match the heterogeneity of the target market (individual clinicians, hospitals of varying size, and occasional pay-per-use).

Table 3: Pricing tiers in Algerian Dinar.

Tier	Customer	Annual price (DZD)
Individual healthcare professional	Per clinician	180,000 to 525,000
Small hospital subscription	Single department	1.5 to 3 million
Medium hospital subscription	Multi-department	3 to 7 million
Large hospital network	Network or group	7 to 15 million
Per-report micro-billing	Occasional or tele-medicine	Approximately 600 DZD per combined report

1.9 Team Development

Team build-up follows the funding phases. Two stages are anticipated:

1.9.1 Initial stage (Phase 1)

The startup begins with a small founding team focused on prototype delivery. Estimated annual budget: 3 to 6 million DZD.

- ▶ Founder and Project Lead (full-stack engineering and product direction).
- ▶ 1 AI / ML Specialist (model integration, calibration, fine-tuning).

1.9.2 Growth stage (Phases 2 and 3)

As the company moves from prototype to pilot deployments and commercial launch, the team expands to seven core roles. Estimated annual payroll budget: 15 to 30 million DZD.

- ▶ 4 Full-Stack Developers (frontend, backend, integration, DevOps).
- ▶ 1 AI / ML Engineer (model maintenance, monitoring and continuous improvement).
- ▶ 1 Clinical Advisor (clinical workflow expert and validation lead).
- ▶ 1 Business Development Specialist (hospital sales and partnerships).

Synthesis

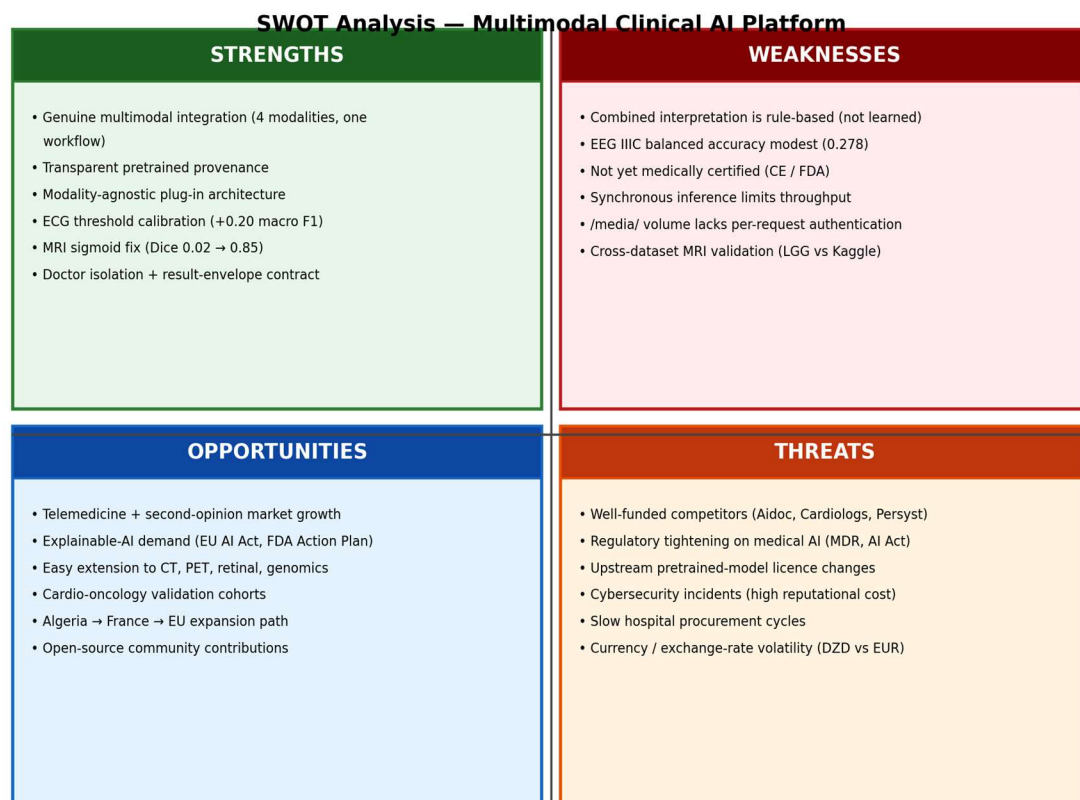


Figure 4: SWOT matrix synthesising the strategic position.

The assumptions above complete the Business Model Canvas with the four blocks that the thesis chapter leaves implicit (Channels, Customer Relationships, Competitive Positioning and Intellectual Property Strategy) and provide concrete numerical anchors (funding targets, pricing tiers, team build-up) expressed in Algerian Dinar. They are consistent with the staged commercialisation roadmap of Chapter 4 and form the starting point for an investor pitch deck.